# Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 20, 2026

## 1 Overview

This document compiles every current SCM case report under results/ that has a compiled wrapper PDF.

# 2 Case 1: sheba 160x2m 12h report

# 3 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

## 3.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 271 / 31
- RHS evaluations: 133040
- Solver tolerances: abstol = $1.000 \times 10^{-8}$, reltol = $1.000 \times 10^{-6}$

## 3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

| **Parameter** | **Symbol** | **Value** |
|---|---|---|
| Zonal geostrophic wind | $U_g$ | $7.000000\,\mathrm{m\,s^{-1}}$ |
| Meridional geostrophic wind | $V_g$ | $0.000000\,\mathrm{m\,s^{-1}}$ |
| Surface roughness (momentum) | $z_{0m}$ | 1.000e-04 m |
| Surface roughness (heat) | $z_{0h}$ | 1.000e-05 m |
| Coriolis parameter | $f$ | 1.000e-04 $\mathrm{s^{-1}}$ |
| Reference potential temperature | $\theta_a$ | 257.000000 K |
| Deep soil temperature | $T_{deep}$ | 255.500000 K |
| Downward longwave forcing | $R_{\downarrow}$ | $190.000000\,\mathrm{W\,m^{-2}}$ |
| Soil conductivity proxy | $\lambda_s$ | 1.200000 |
| Soil slab depth | $d_{soil}$ | 0.100000 m |
| Surface diffusivity floor | $k_{\min,\mathrm{surf}}$ | $0.001000\,\mathrm{m^2\,s^{-1}}$ |
| Top thermal BC type | — | neumann |
| Top boundary potential temperature | $\theta_{top}$ | 257.000000 K |
| Top relaxation coefficient | $\lambda_{top}$ | $0.000000\,\mathrm{s^{-1}}$ |

## 3.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.023325,\ 6.644851]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.030422,\ 1870.864769]$

### 3.4 Generated Artifacts

- Time-series CSV: `results/sheba/time_series.csv`
- Diagnostic summary JSON: `results/sheba/summary.json`
- Payload JLD2: `results/sheba/payload.jld2`

### 3.5 Figure Manifest

1. Time series of T__s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

### 3.6 Included Plots

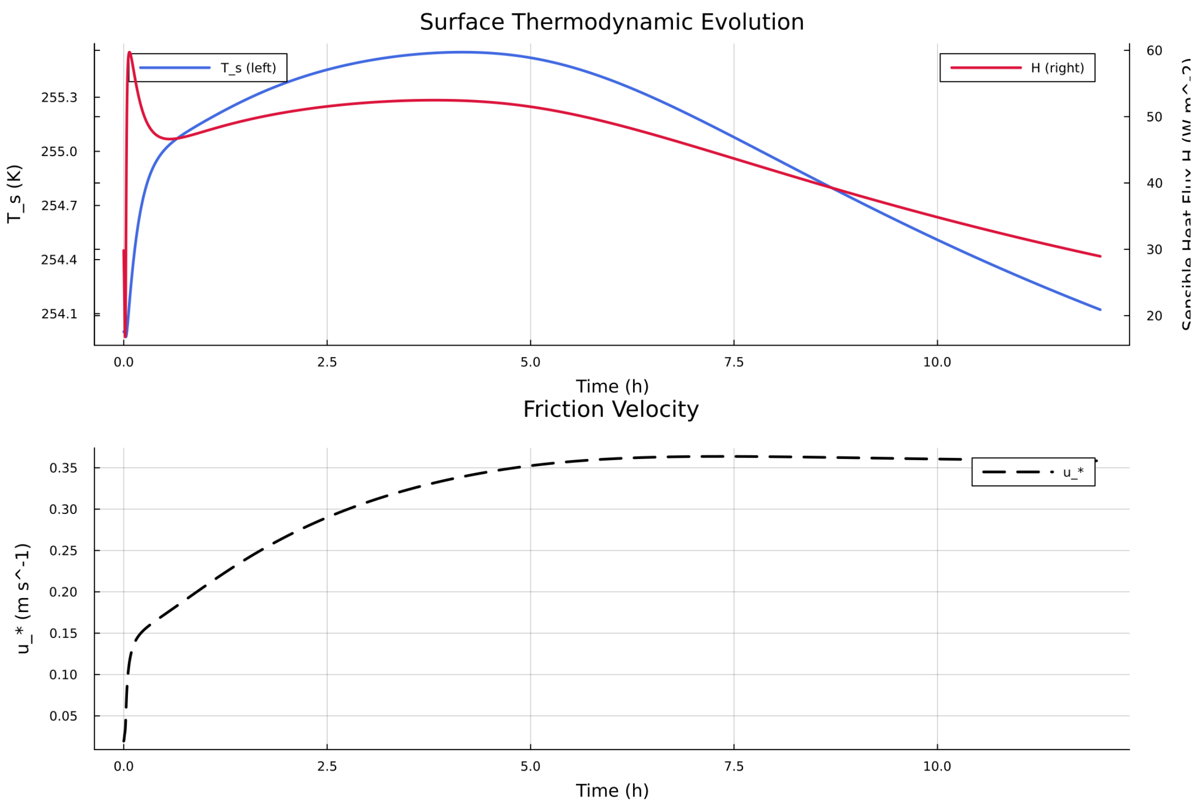


Figure 1: Time series of T__s, sensible heat flux, and friction velocity

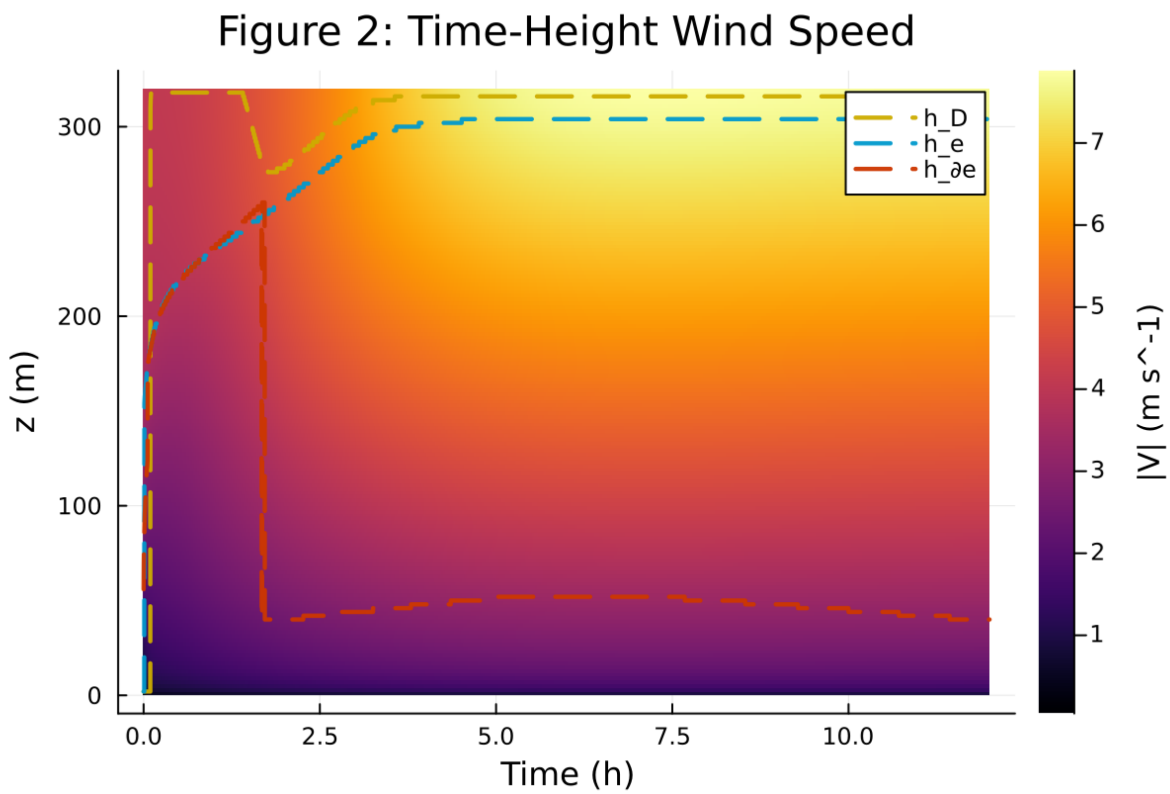


Figure 2: Time-height contour of wind speed with LLJ evolution

# 4 Case 2: gabls1 80x2m 9h report

# 5 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

## 5.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 90 / 0
- RHS evaluations: 22502
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

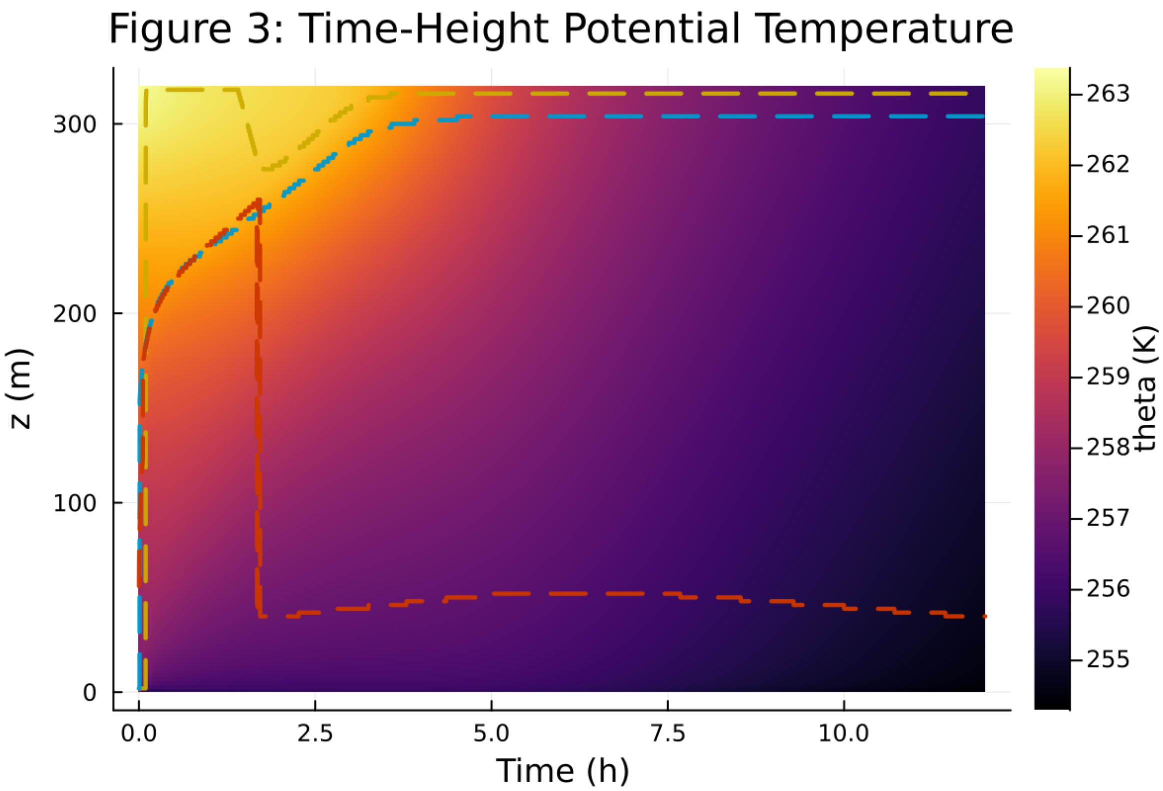


Figure 3: Time-height contour of potential temperature and inversion growth

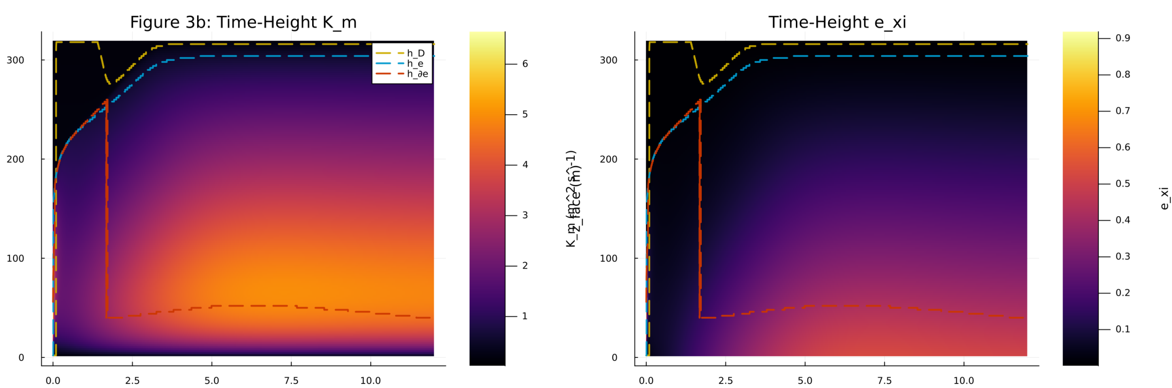


Figure 4: Vertical profiles of U, theta, and Km at representative times

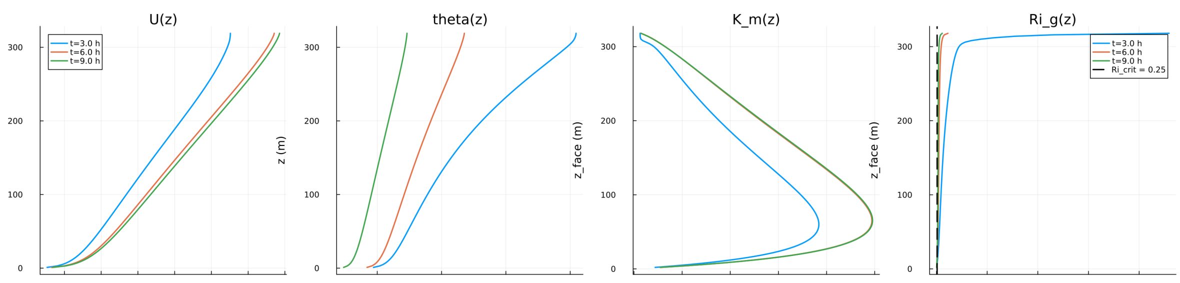


Figure 5: Surface energy budget components (R__n, H, G, storage)

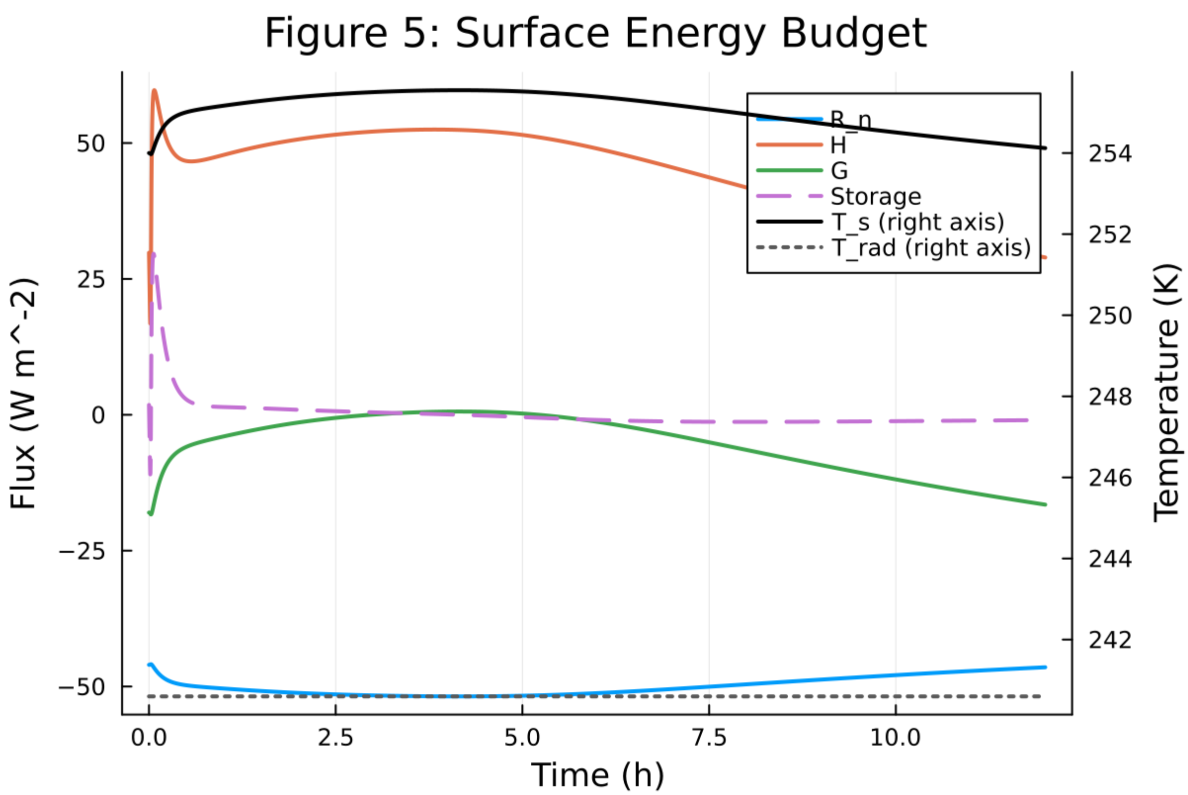


Figure 6: Phase portrait on regularized manifold (Delta vs e_xi)

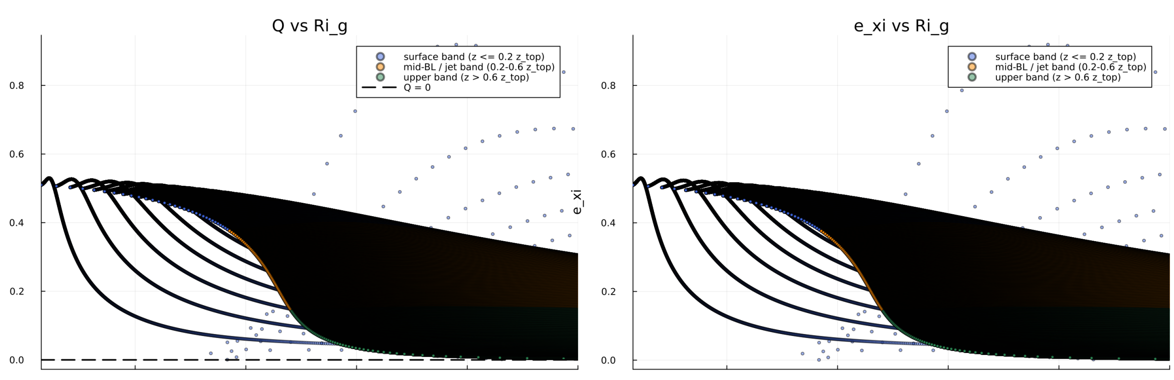


Figure 7: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

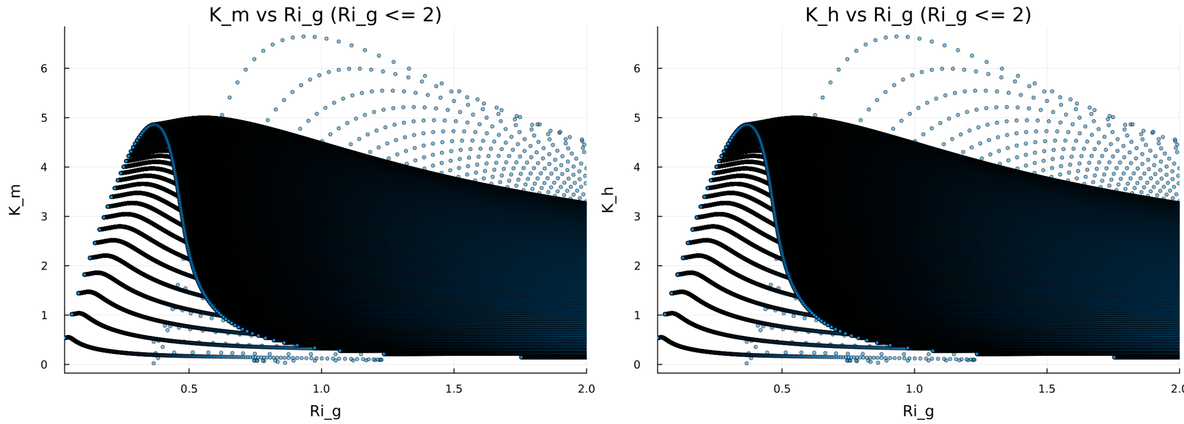


Figure 8: Fold proximity diagnostics versus time

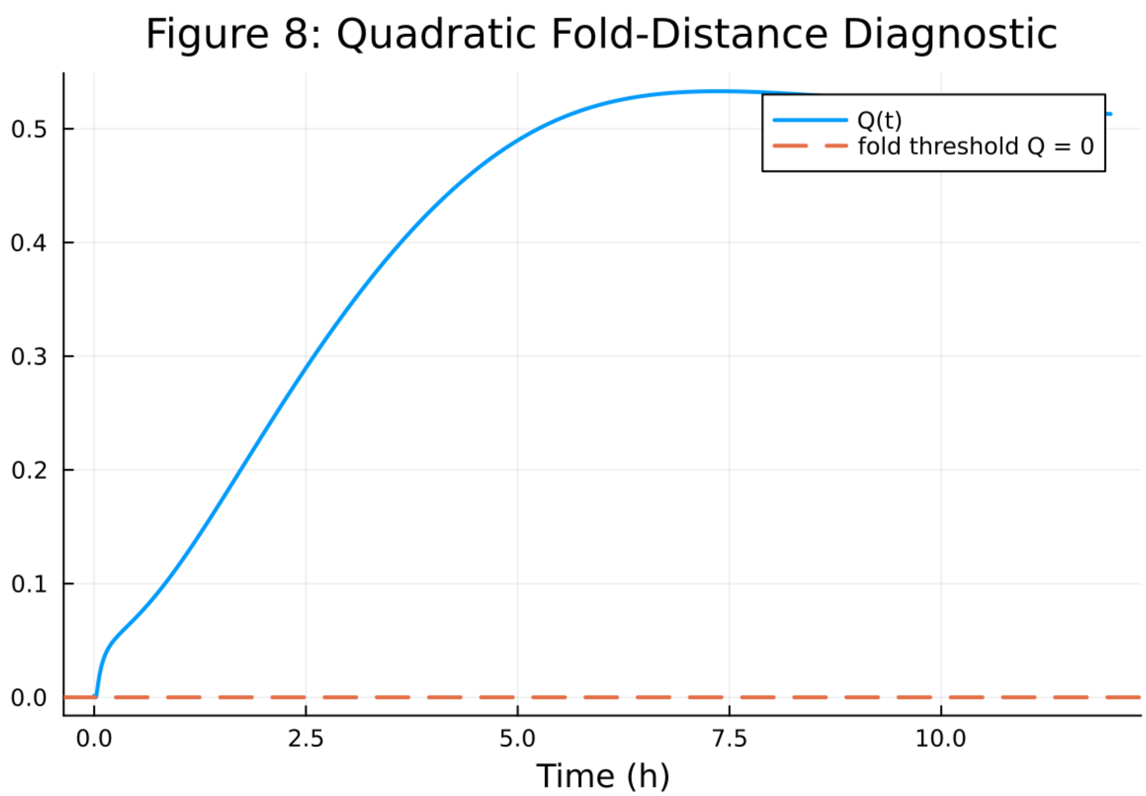


Figure 9: fig08 fold proximity.png

Table 2: Forcing and Boundary Condition Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Zonal geostrophic wind | $U_g$ | 8.000000 $\mathrm{m\,s^{-1}}$ |
| Meridional geostrophic wind | $V_g$ | 0.000000 $\mathrm{m\,s^{-1}}$ |
| Surface roughness (momentum) | $z_{0m}$ | 0.100000 m |
| Surface roughness (heat) | $z_{0h}$ | 0.010000 m |
| Coriolis parameter | $f$ | 1.000e-04 $\mathrm{s^{-1}}$ |
| Reference potential temperature | $\theta_a$ | 265.000000 K |
| Deep soil temperature | $T_{deep}$ | 262.000000 K |
| Downward longwave forcing | $R_{\downarrow}$ | 245.000000 $\mathrm{W\,m^{-2}}$ |
| Soil conductivity proxy | $\lambda_s$ | 1.200000 |
| Soil slab depth | $d_{soil}$ | 0.100000 m |
| Surface diffusivity floor | $k_{\mathrm{min,surf}}$ | 0.001000 $\mathrm{m^2\,s^{-1}}$ |
| Top thermal BC type | — | neumann |
| Top boundary potential temperature | $\theta_{top}$ | 265.000000 K |
| Top relaxation coefficient | $\lambda_{top}$ | 0.000000 $\mathrm{s^{-1}}$ |

## 5.2 Forcing and Boundary Conditions

## 5.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.023104, 3.564385]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.012198, 4.959639]$

## 5.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

## 5.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

## 5.6 Included Plots

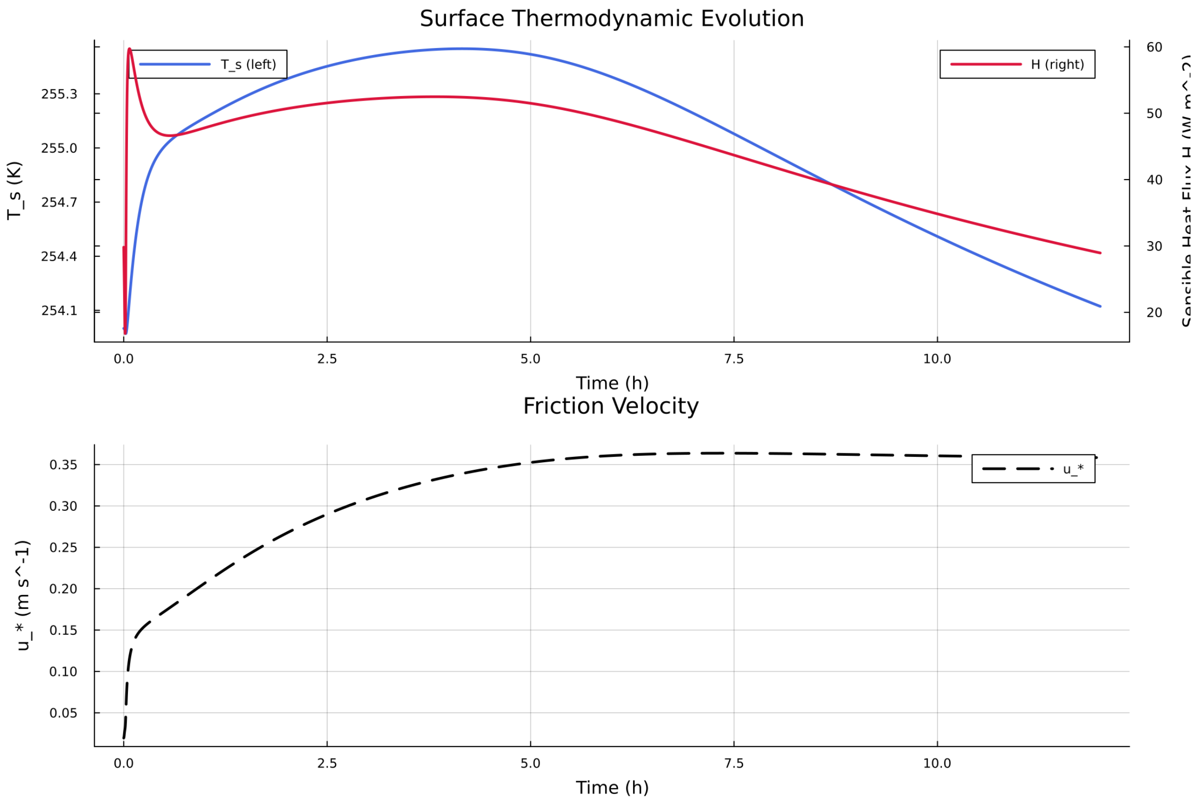


Figure 10: Time series of T__s, sensible heat flux, and friction velocity

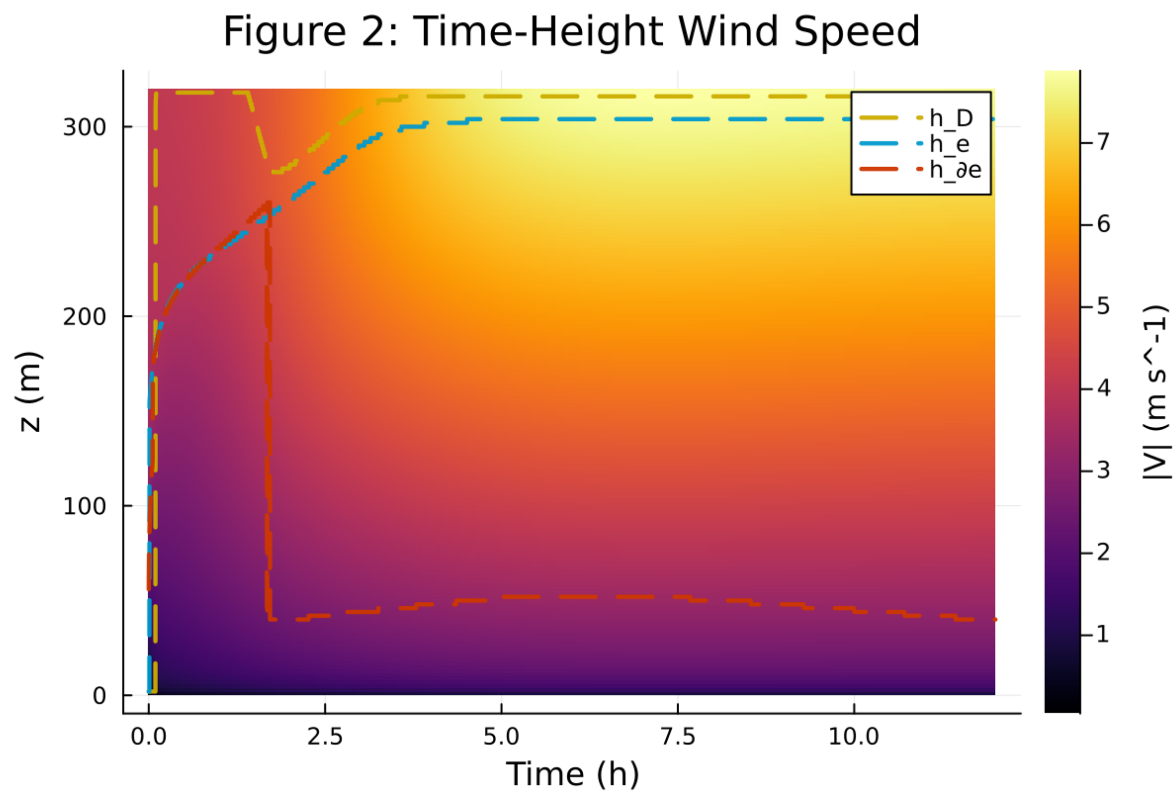


Figure 11: Time-height contour of wind speed with LLJ evolution

# 6 Case 3: idealized sbl 80x2m 9h report

# 7 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

## 7.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 106 / 0
- RHS evaluations: 26608
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

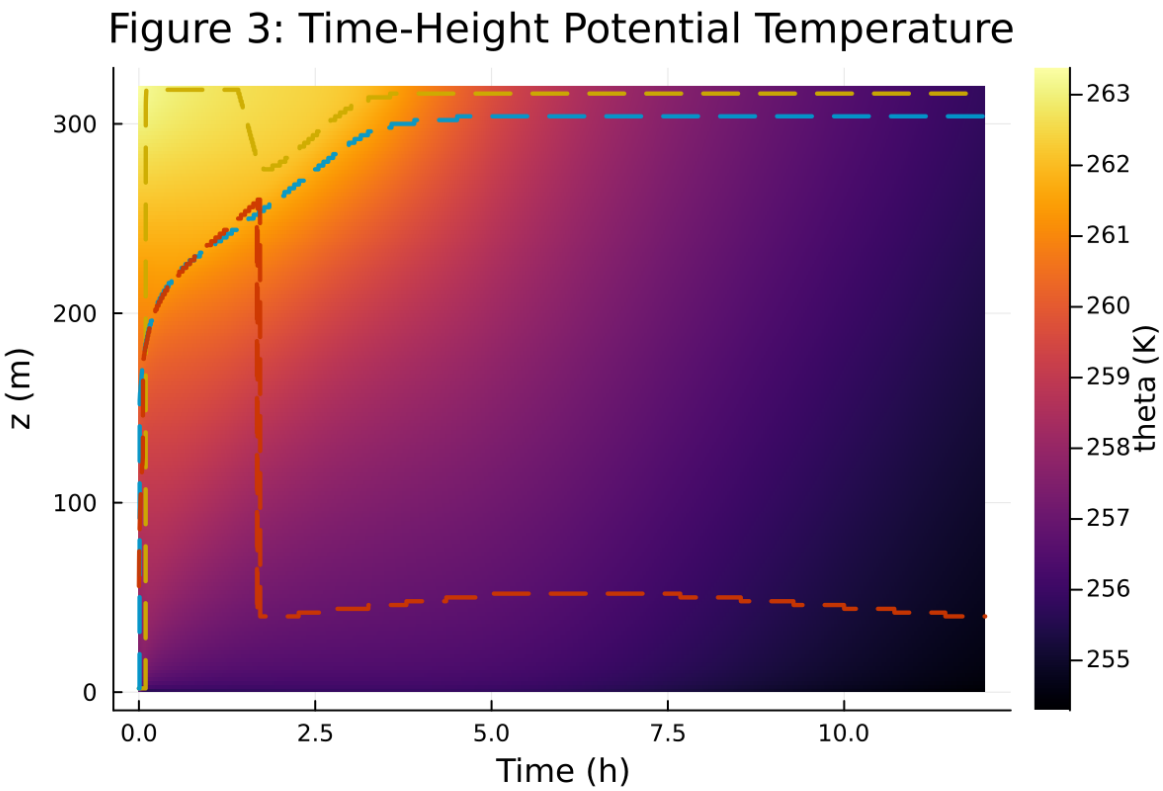


Figure 12: Time-height contour of potential temperature and inversion growth

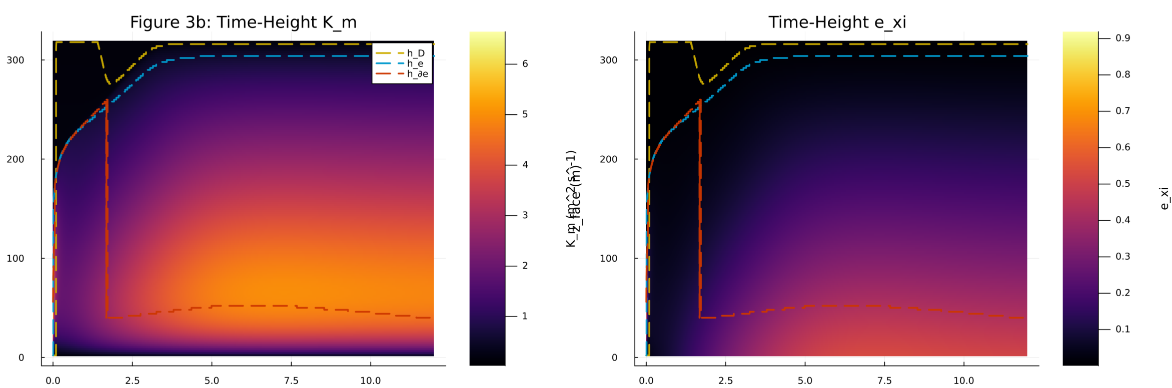


Figure 13: Vertical profiles of U, theta, and Km at representative times

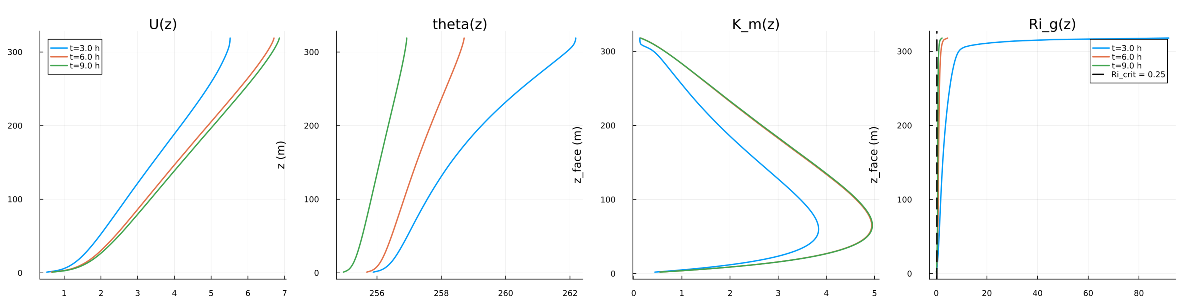


Figure 14: Surface energy budget components (R_n, H, G, storage)

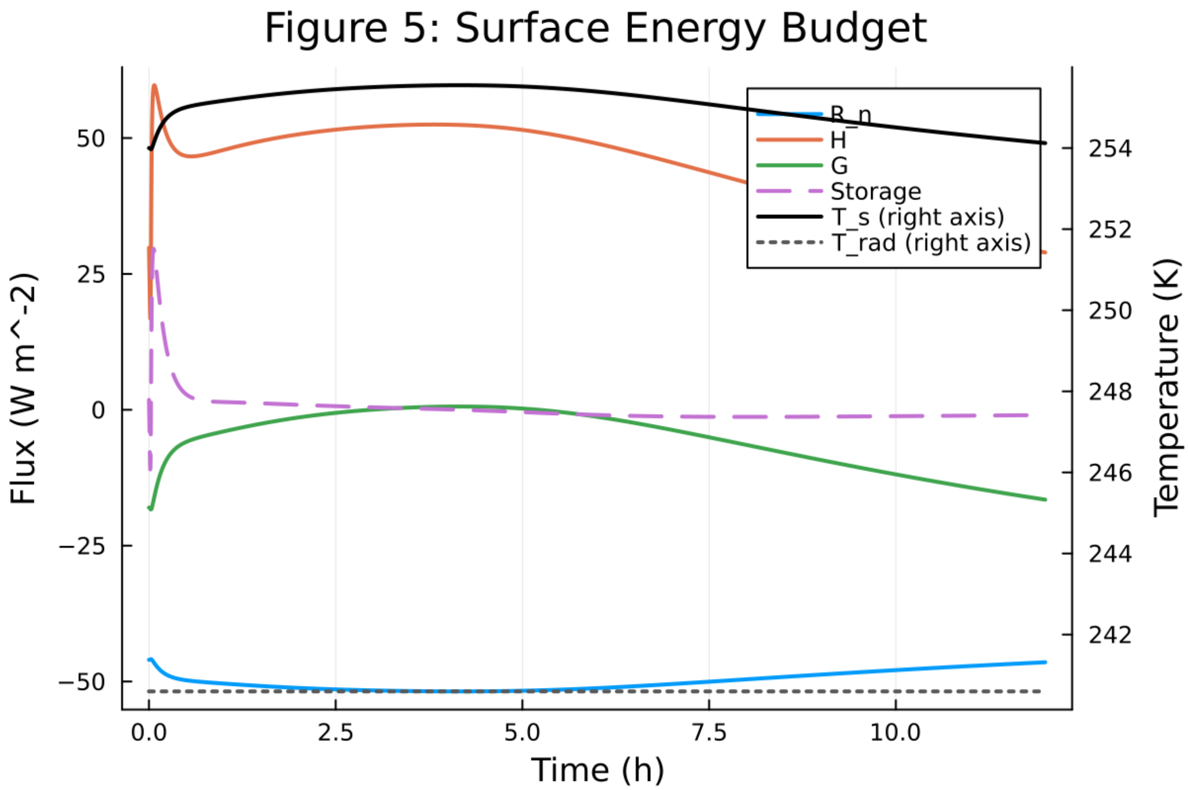


Figure 15: Phase portrait on regularized manifold (Delta vs e__xi)

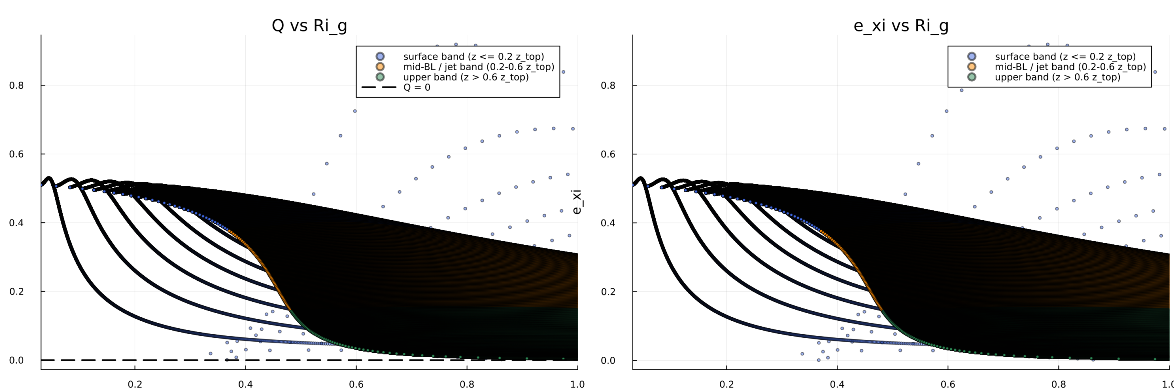


Figure 16: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

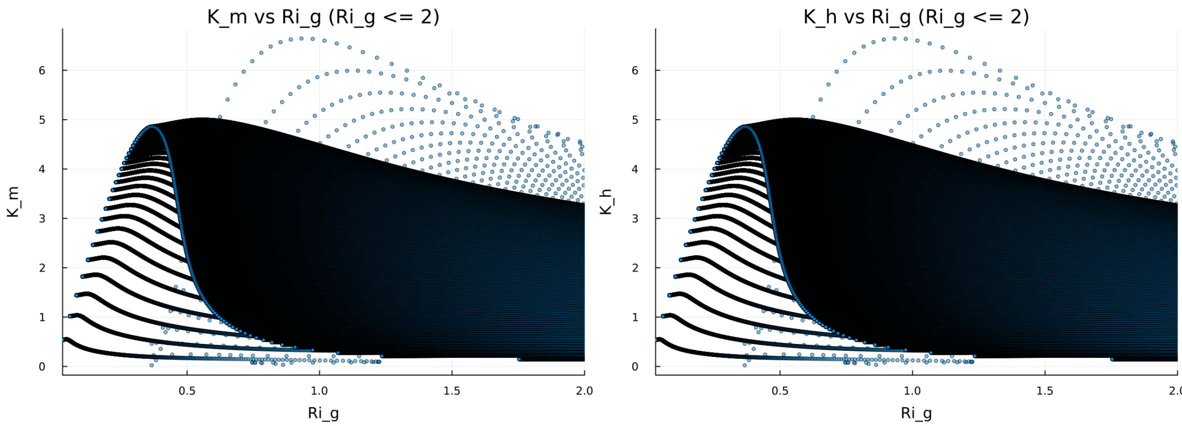


Figure 17: Fold proximity diagnostics versus time

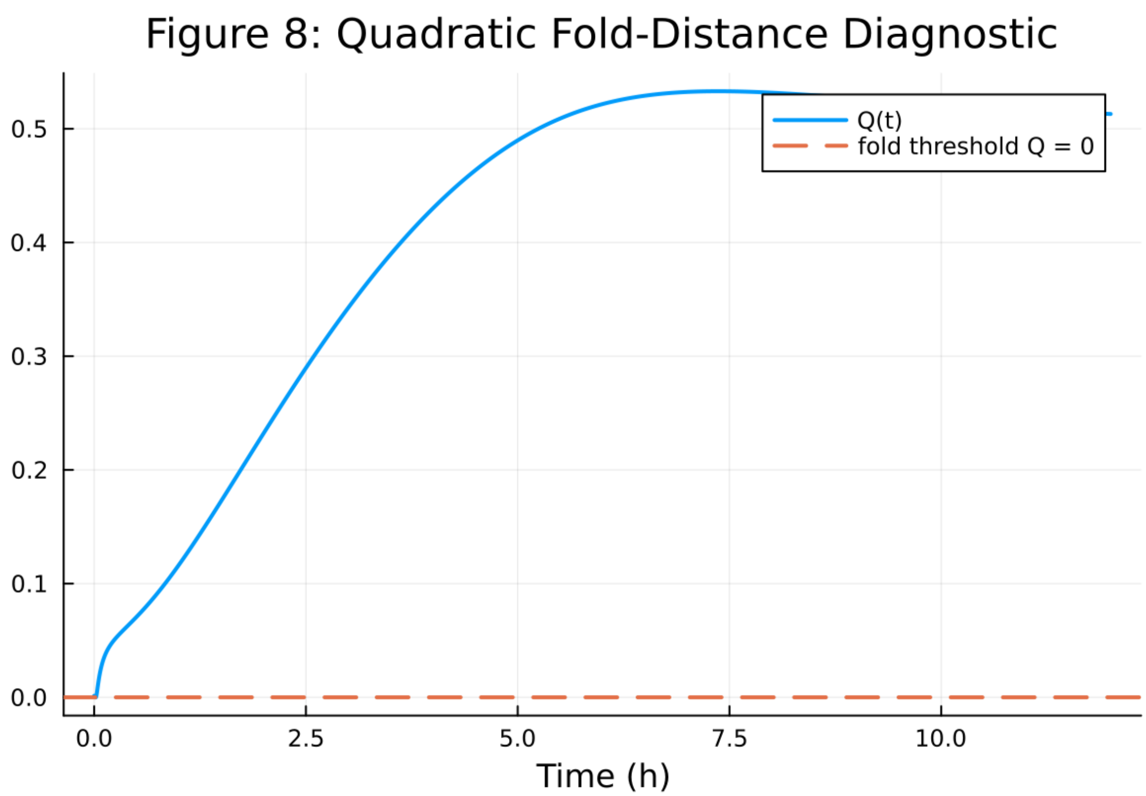


Figure 18: fig08 fold proximity.png

Table 3: Forcing and Boundary Condition Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Zonal geostrophic wind | $U_g$ | $10.000000\,\mathrm{m\,s^{-1}}$ |
| Meridional geostrophic wind | $V_g$ | $0.000000\,\mathrm{m\,s^{-1}}$ |
| Surface roughness (momentum) | $z_{0m}$ | $0.020000\,\mathrm{m}$ |
| Surface roughness (heat) | $z_{0h}$ | $0.005000\,\mathrm{m}$ |
| Coriolis parameter | $f$ | $1.000\text{e-}04\,\mathrm{s^{-1}}$ |
| Reference potential temperature | $\theta_a$ | $280.000000\,\mathrm{K}$ |
| Deep soil temperature | $T_{deep}$ | $276.000000\,\mathrm{K}$ |
| Downward longwave forcing | $R_{\downarrow}$ | $300.000000\,\mathrm{W\,m^{-2}}$ |
| Soil conductivity proxy | $\lambda_s$ | 1.200000 |
| Soil slab depth | $d_{soil}$ | $0.100000\,\mathrm{m}$ |
| Surface diffusivity floor | $k_{\mathrm{min,surf}}$ | $0.001000\,\mathrm{m^2\,s^{-1}}$ |
| Top thermal BC type | — | neumann |
| Top boundary potential temperature | $\theta_{top}$ | $265.000000\,\mathrm{K}$ |
| Top relaxation coefficient | $\lambda_{top}$ | $0.000000\,\mathrm{s^{-1}}$ |

### 7.2 Forcing and Boundary Conditions

### 7.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.057081,\ 9.568304]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.009944,\ 1.655811]$

### 7.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

### 7.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

### 7.6 Included Plots

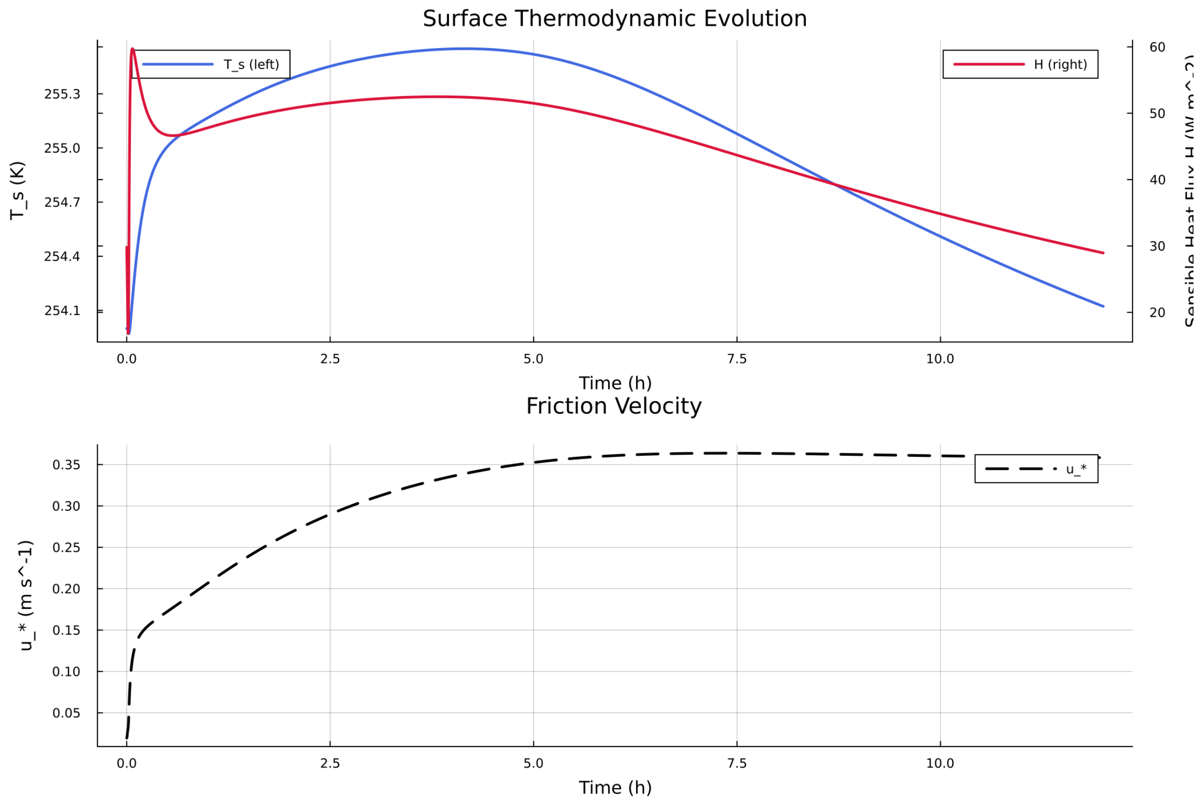


Figure 19: Time series of T__s, sensible heat flux, and friction velocity

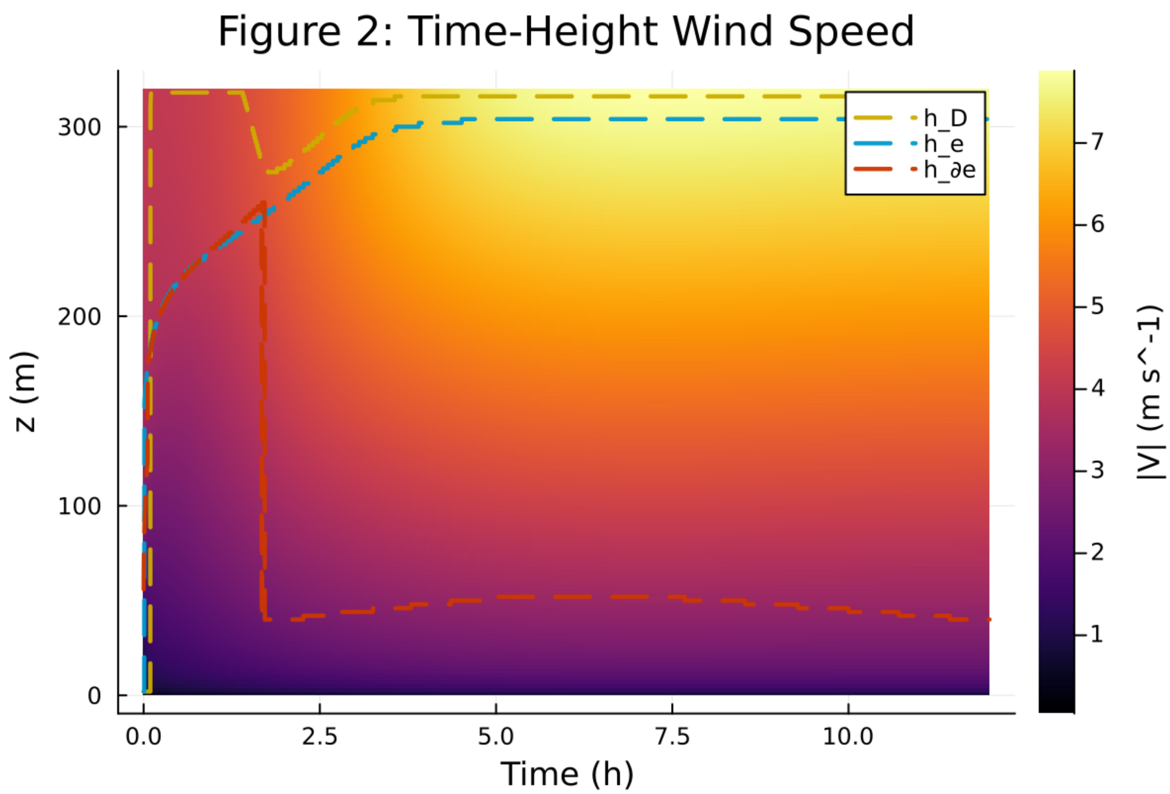


Figure 20: Time-height contour of wind speed with LLJ evolution

# 8 Case 4: sheba 160x2m 12h report

# 9 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba_fd/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

## 9.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba_fd`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 273 / 33
- RHS evaluations: 134309
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

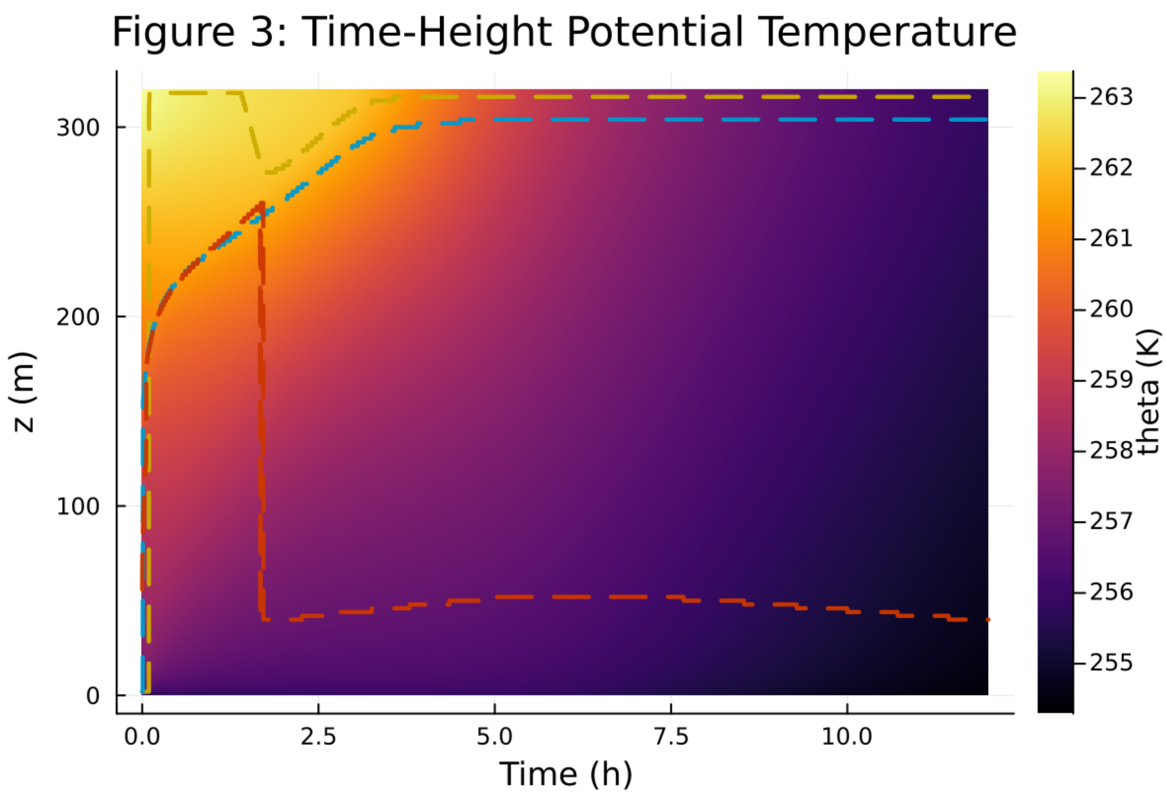


Figure 21: Time-height contour of potential temperature and inversion growth

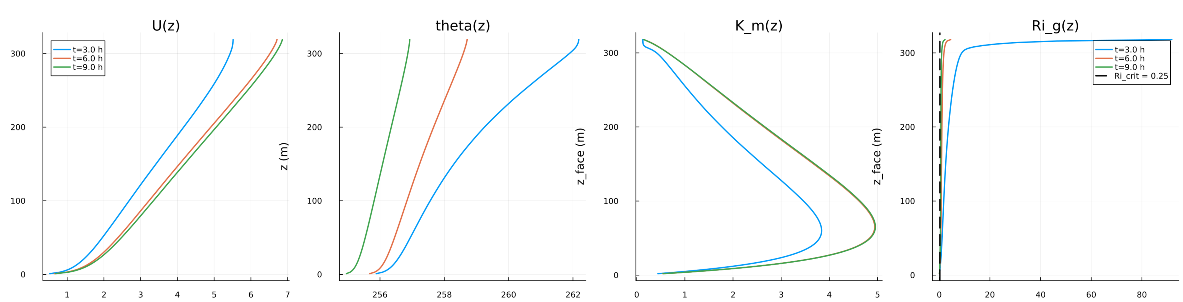


Figure 22: Vertical profiles of U, theta, and Km at representative times

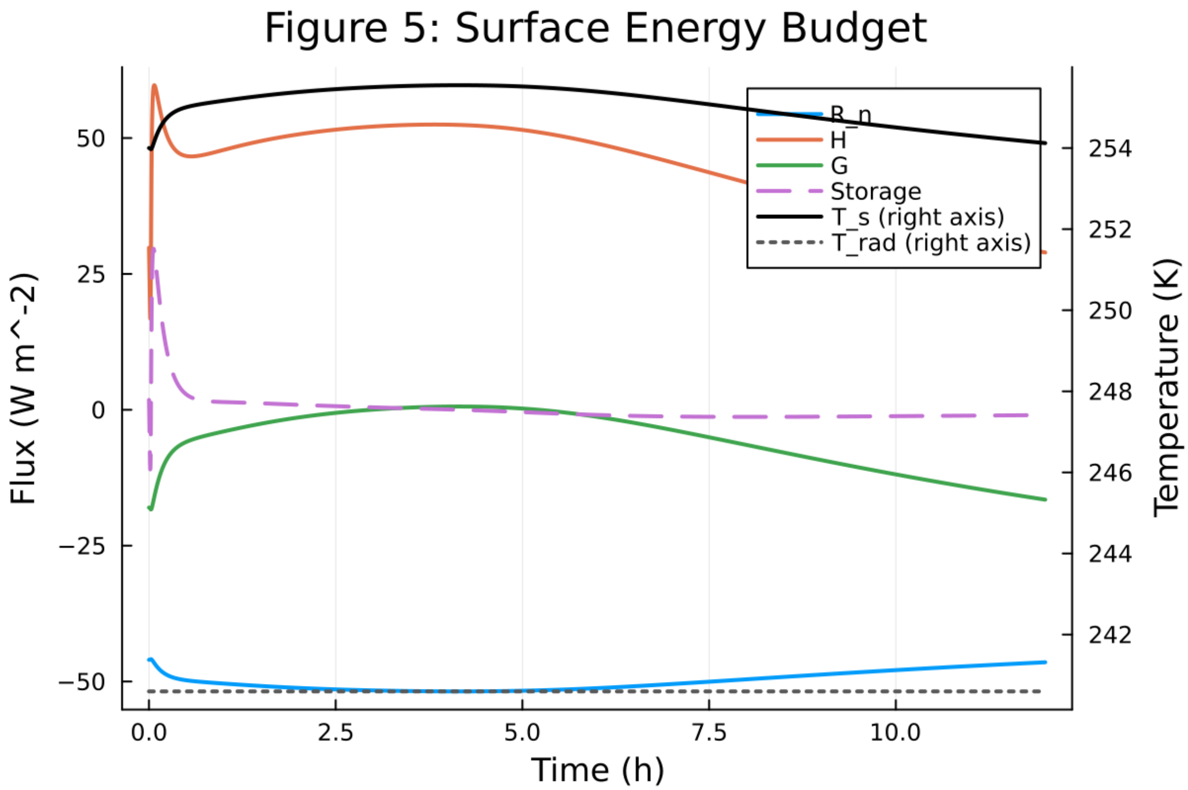


Figure 23: Surface energy budget components (R__n, H, G, storage)

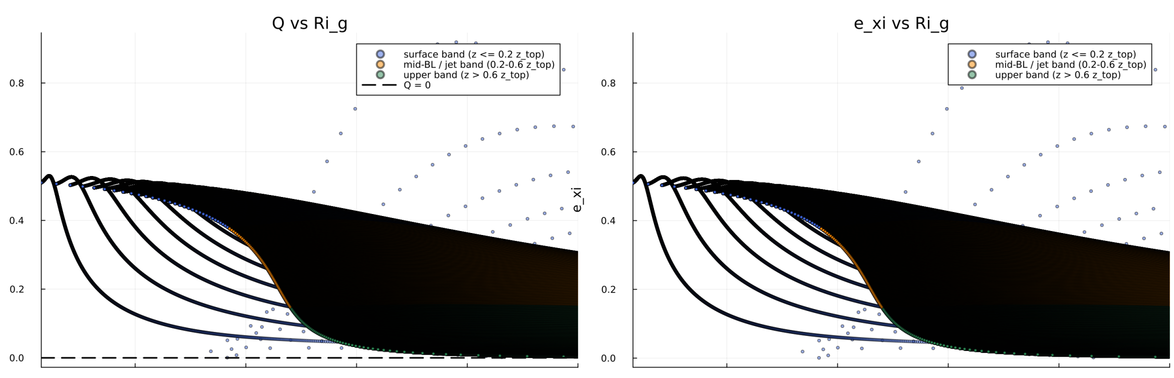


Figure 24: Phase portrait on regularized manifold (Delta vs e__xi)

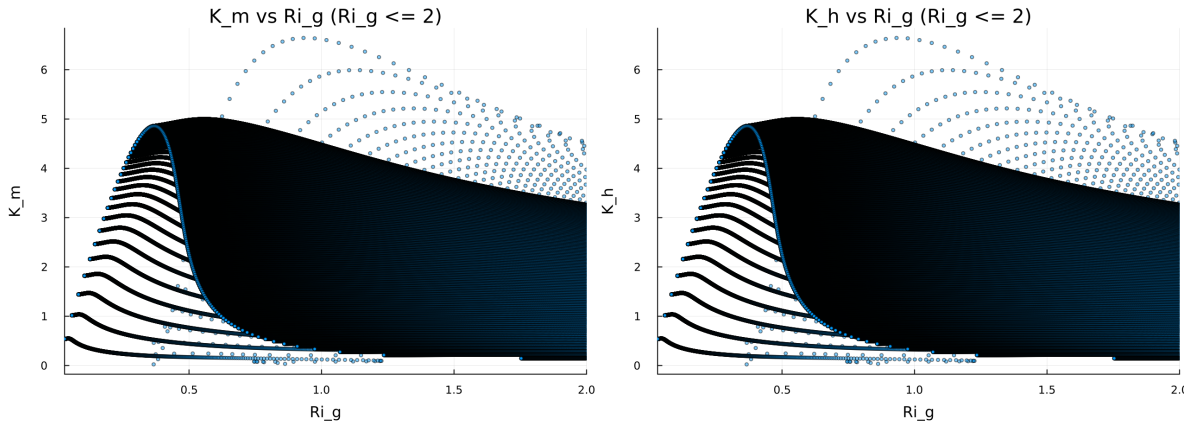


Figure 25: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

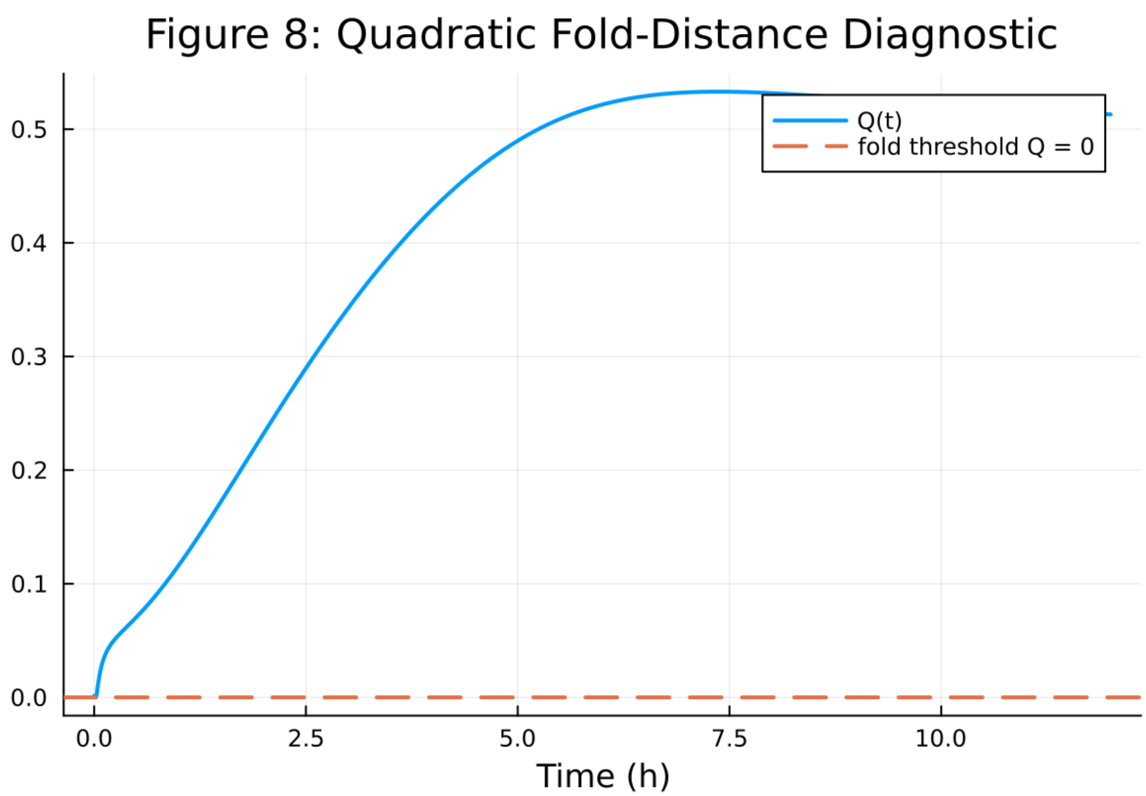


Figure 26: Fold proximity diagnostics versus time

Table 4: Forcing and Boundary Condition Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Zonal geostrophic wind | $U_g$ | 7.000000 $\mathrm{m\,s^{-1}}$ |
| Meridional geostrophic wind | $V_g$ | 0.000000 $\mathrm{m\,s^{-1}}$ |
| Surface roughness (momentum) | $z_{0m}$ | 1.000e-04 m |
| Surface roughness (heat) | $z_{0h}$ | 1.000e-05 m |
| Coriolis parameter | $f$ | 1.000e-04 $\mathrm{s^{-1}}$ |
| Reference potential temperature | $\theta_a$ | 257.000000 K |
| Deep soil temperature | $T_{deep}$ | 255.500000 K |
| Downward longwave forcing | $R_\downarrow$ | 190.000000 $\mathrm{W\,m^{-2}}$ |
| Soil conductivity proxy | $\lambda_s$ | 1.200000 |
| Soil slab depth | $d_{soil}$ | 0.100000 m |
| Surface diffusivity floor | $k_{\mathrm{min,surf}}$ | 0.001000 $\mathrm{m^2\,s^{-1}}$ |
| Top thermal BC type | — | neumann |
| Top boundary potential temperature | $\theta_{top}$ | 257.000000 K |
| Top relaxation coefficient | $\lambda_{top}$ | 0.000000 $\mathrm{s^{-1}}$ |

## 9.2 Forcing and Boundary Conditions

## 9.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.023325, 6.644851]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.030421, 1870.864717]$

## 9.4 Generated Artifacts

- Time-series CSV: `results/sheba_fd/time_series.csv`
- Diagnostic summary JSON: `results/sheba_fd/summary.json`
- Payload JLD2: `results/sheba_fd/payload.jld2`

## 9.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

## 9.6 Included Plots

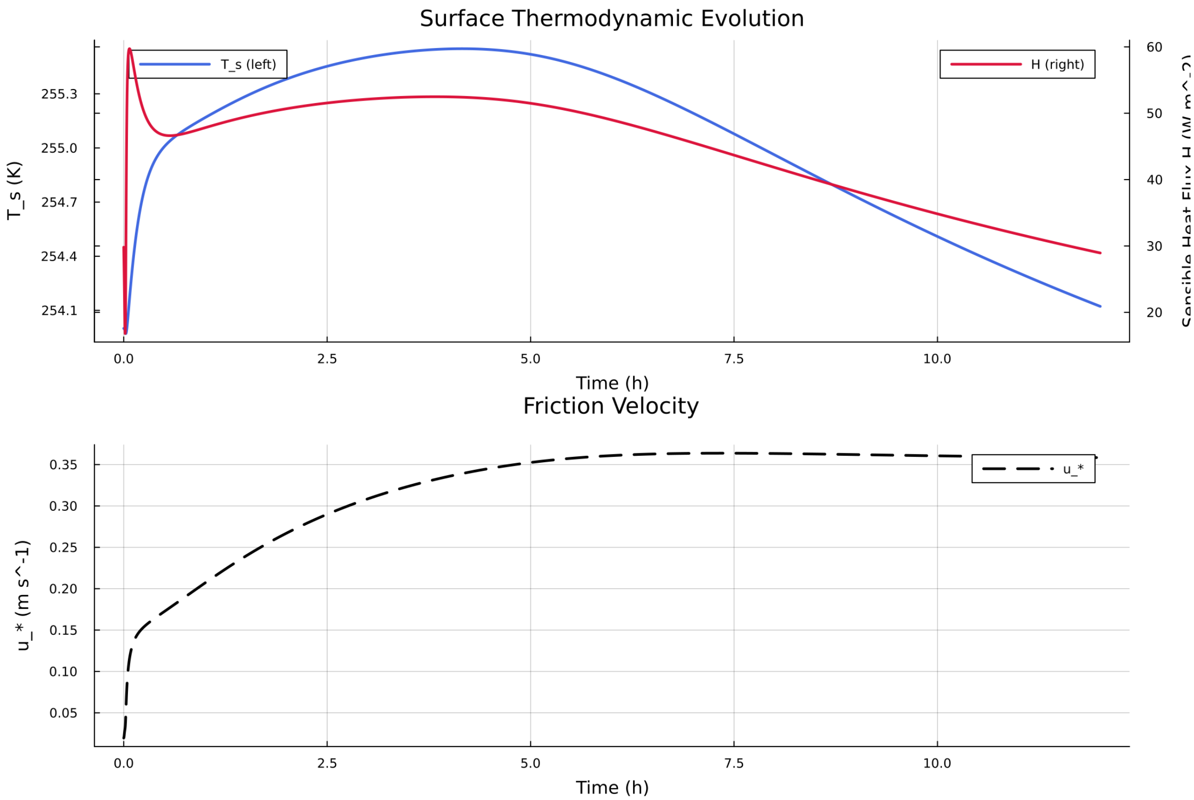


Figure 27: Time series of T__s, sensible heat flux, and friction velocity

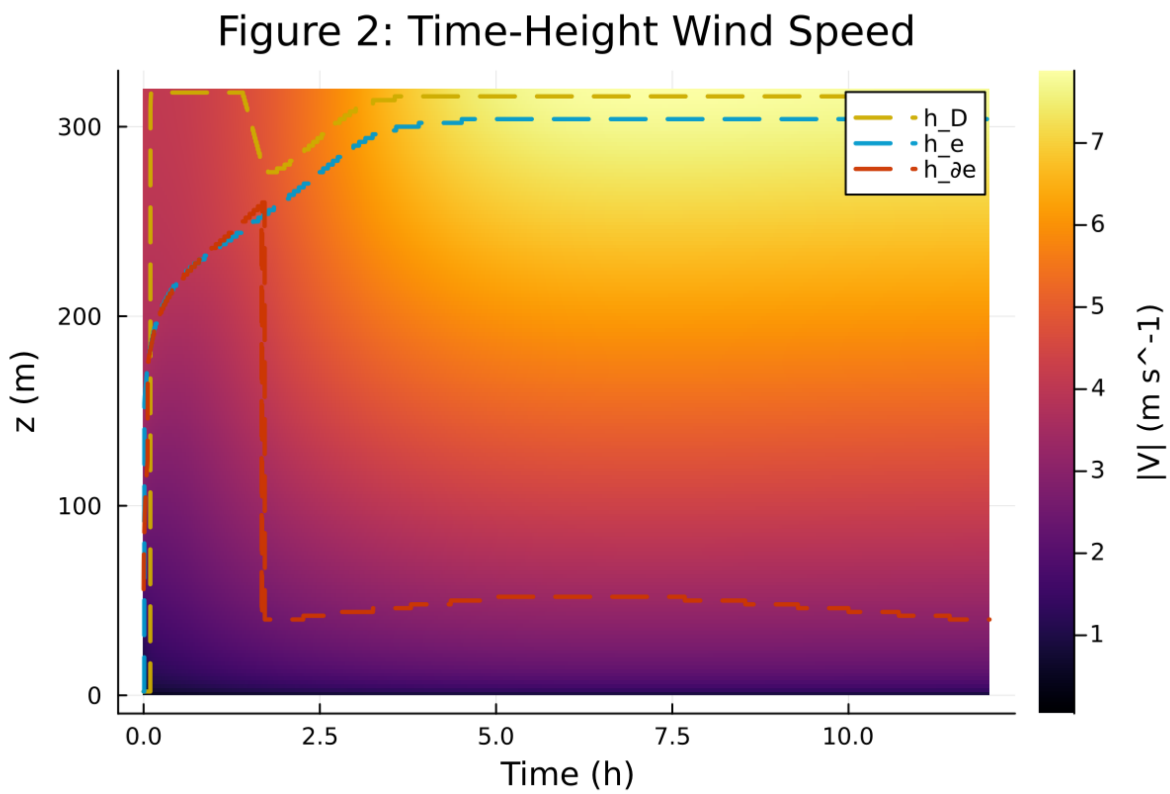


Figure 28: Time-height contour of wind speed with LLJ evolution

# 10 Case 5: sheba 250x2m 12h report

# 11 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba_high_top/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

## 11.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba_high_top`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 151 / 1
- RHS evaluations: 2728
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

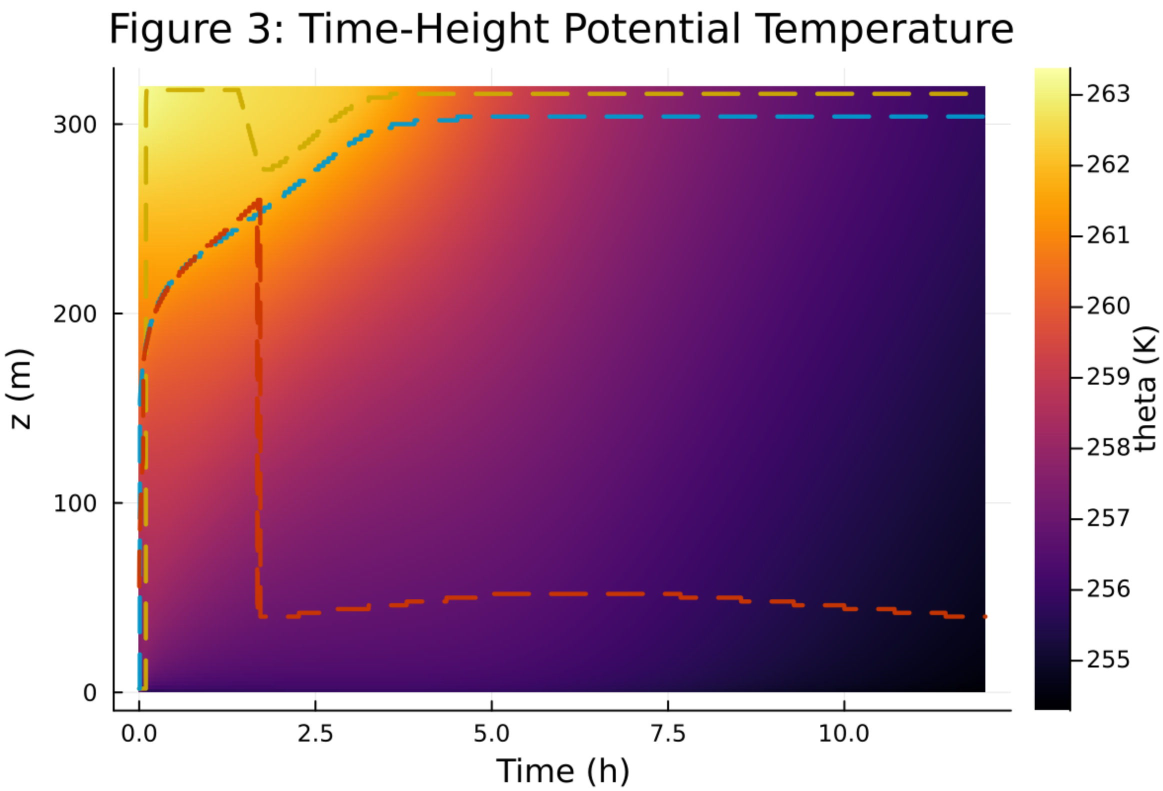


Figure 29: Time-height contour of potential temperature and inversion growth

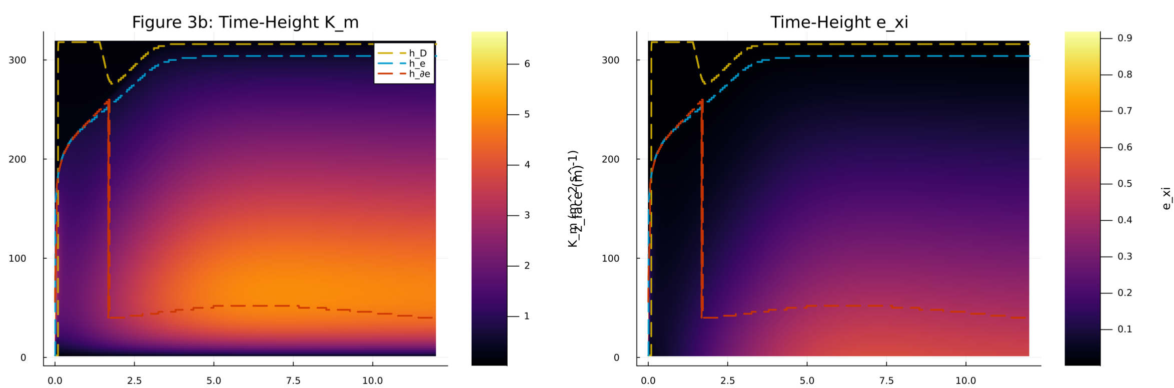


Figure 30: Vertical profiles of U, theta, and Km at representative times

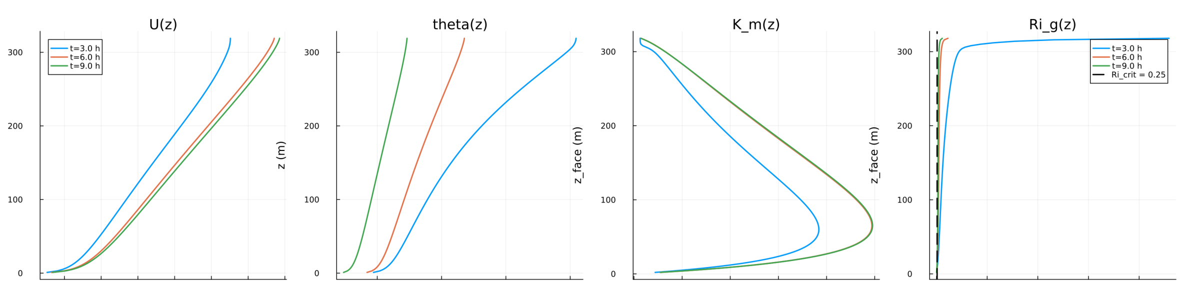


Figure 31: Surface energy budget components (R_n, H, G, storage)

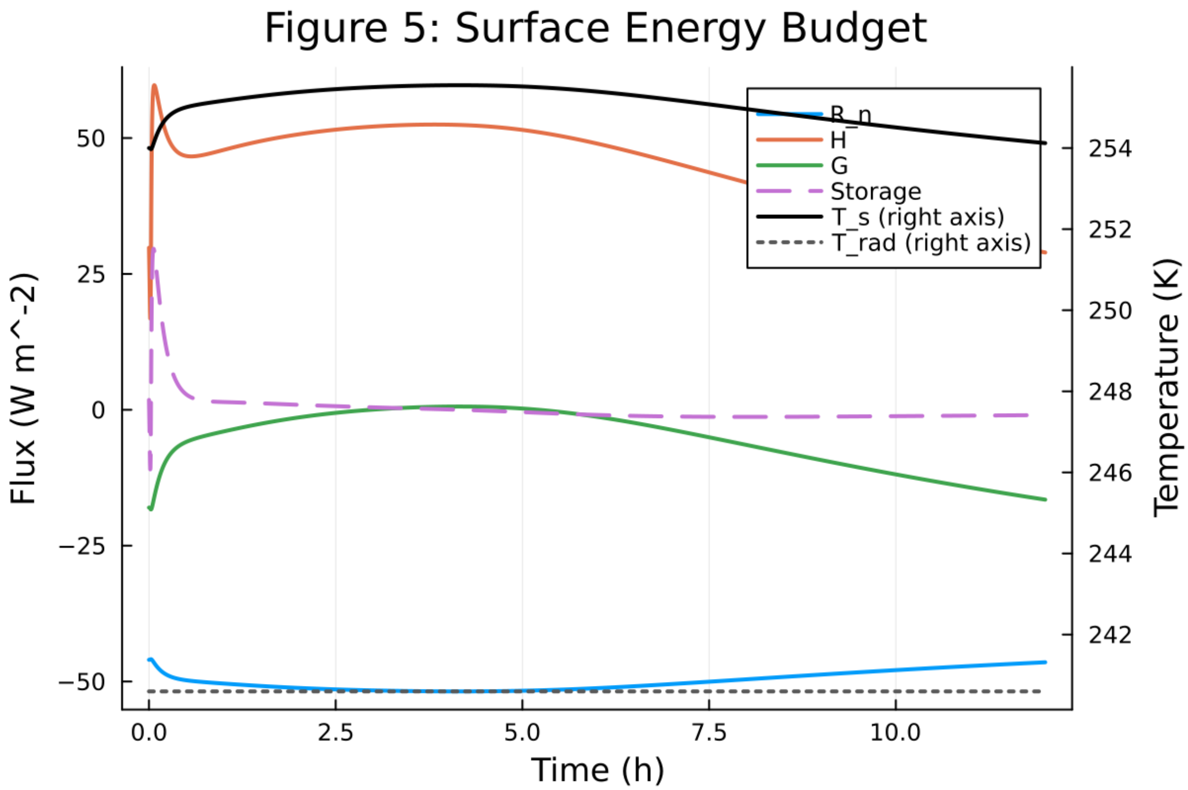


Figure 32: Phase portrait on regularized manifold (Delta vs e__xi)

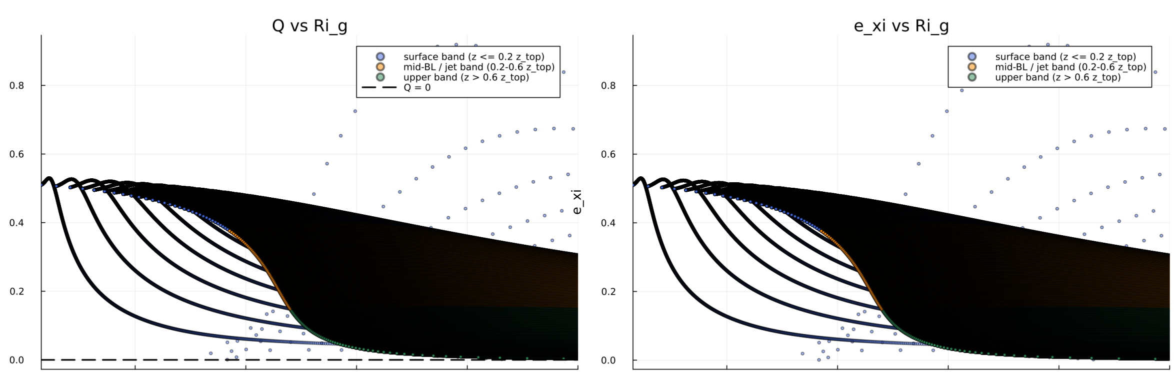


Figure 33: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

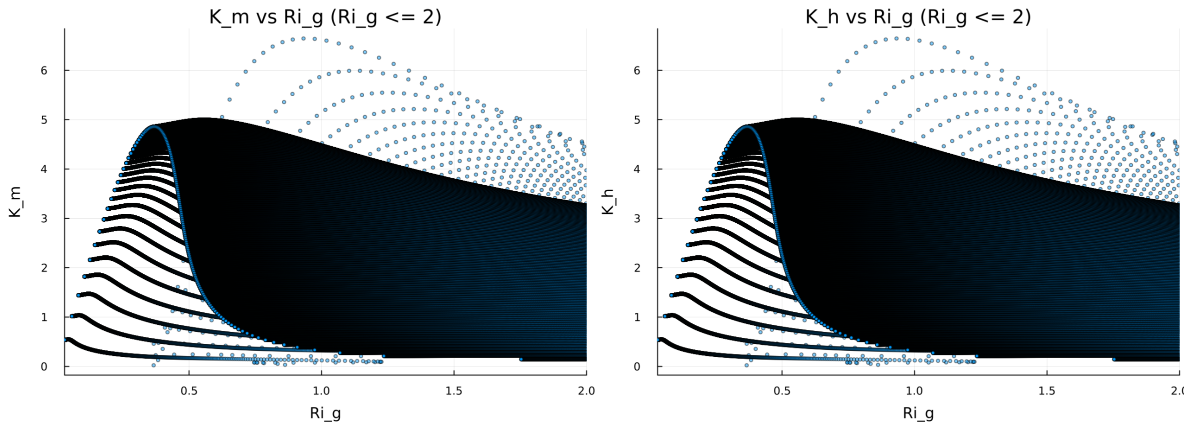


Figure 34: Fold proximity diagnostics versus time

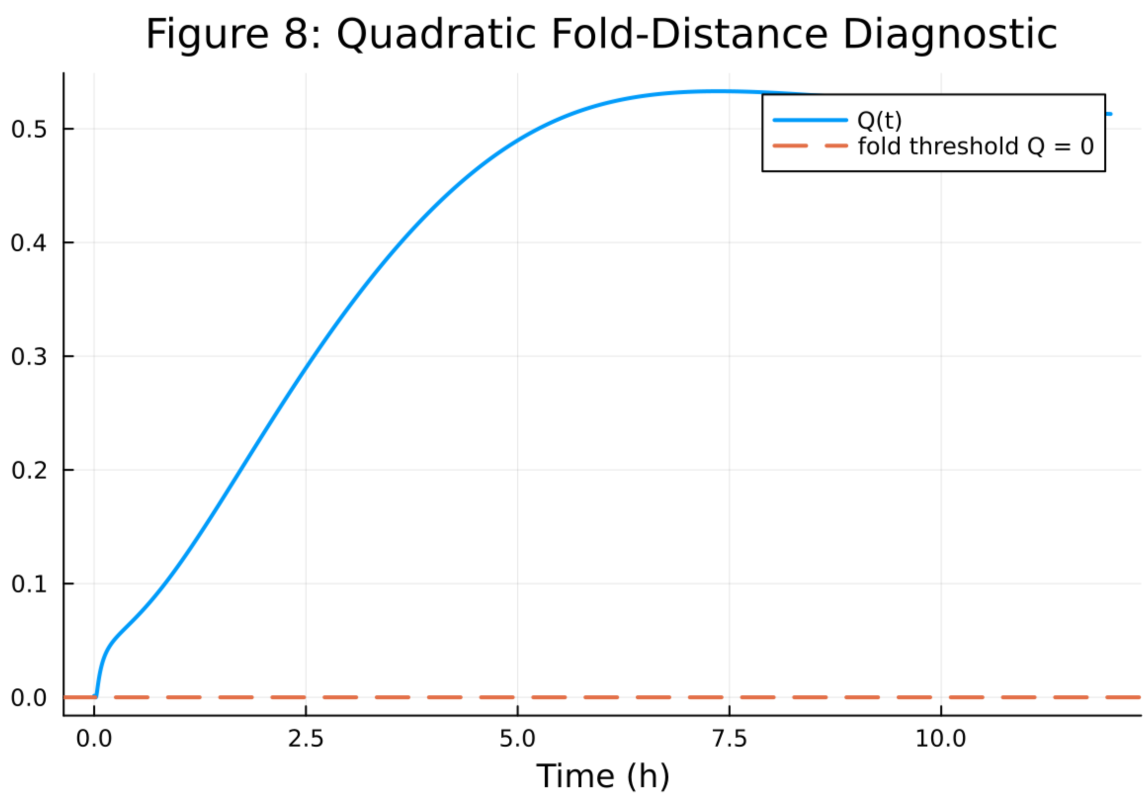


Figure 35: fig08 fold proximity.png

Table 5: Forcing and Boundary Condition Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Zonal geostrophic wind | $U_g$ | 7.000000 $\mathrm{m\,s^{-1}}$ |
| Meridional geostrophic wind | $V_g$ | 0.000000 $\mathrm{m\,s^{-1}}$ |
| Surface roughness (momentum) | $z_{0m}$ | 1.000e-04 m |
| Surface roughness (heat) | $z_{0h}$ | 1.000e-05 m |
| Coriolis parameter | $f$ | 1.000e-04 $\mathrm{s^{-1}}$ |
| Reference potential temperature | $\theta_a$ | 257.000000 K |
| Deep soil temperature | $T_{deep}$ | 255.500000 K |
| Downward longwave forcing | $R_\downarrow$ | 190.000000 $\mathrm{W\,m^{-2}}$ |
| Soil conductivity proxy | $\lambda_s$ | 1.200000 |
| Soil slab depth | $d_{soil}$ | 0.100000 m |
| Surface diffusivity floor | $k_{\mathrm{min,surf}}$ | 0.001000 $\mathrm{m^2\,s^{-1}}$ |
| Top thermal BC type | — | neumann |
| Top boundary potential temperature | $\theta_{top}$ | 257.000000 K |
| Top relaxation coefficient | $\lambda_{top}$ | 0.000000 $\mathrm{s^{-1}}$ |

## 11.2 Forcing and Boundary Conditions

## 11.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.023796, 7.535015]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.030581, 3.273 \times 10^4]$

## 11.4 Generated Artifacts

- Time-series CSV: `results/sheba_high_top/time_series.csv`
- Diagnostic summary JSON: `results/sheba_high_top/summary.json`
- Payload JLD2: `results/sheba_high_top/payload.jld2`

## 11.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

## 11.6 Included Plots

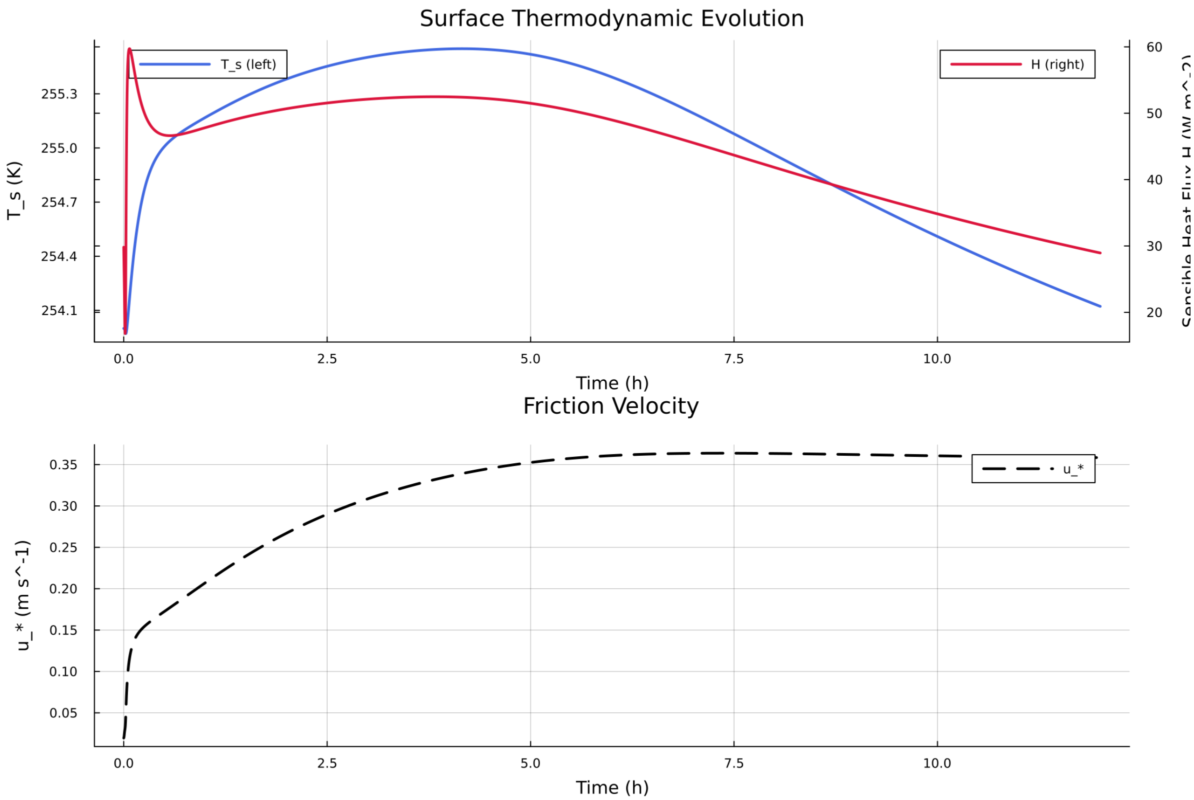


Figure 36: Time series of T__s, sensible heat flux, and friction velocity

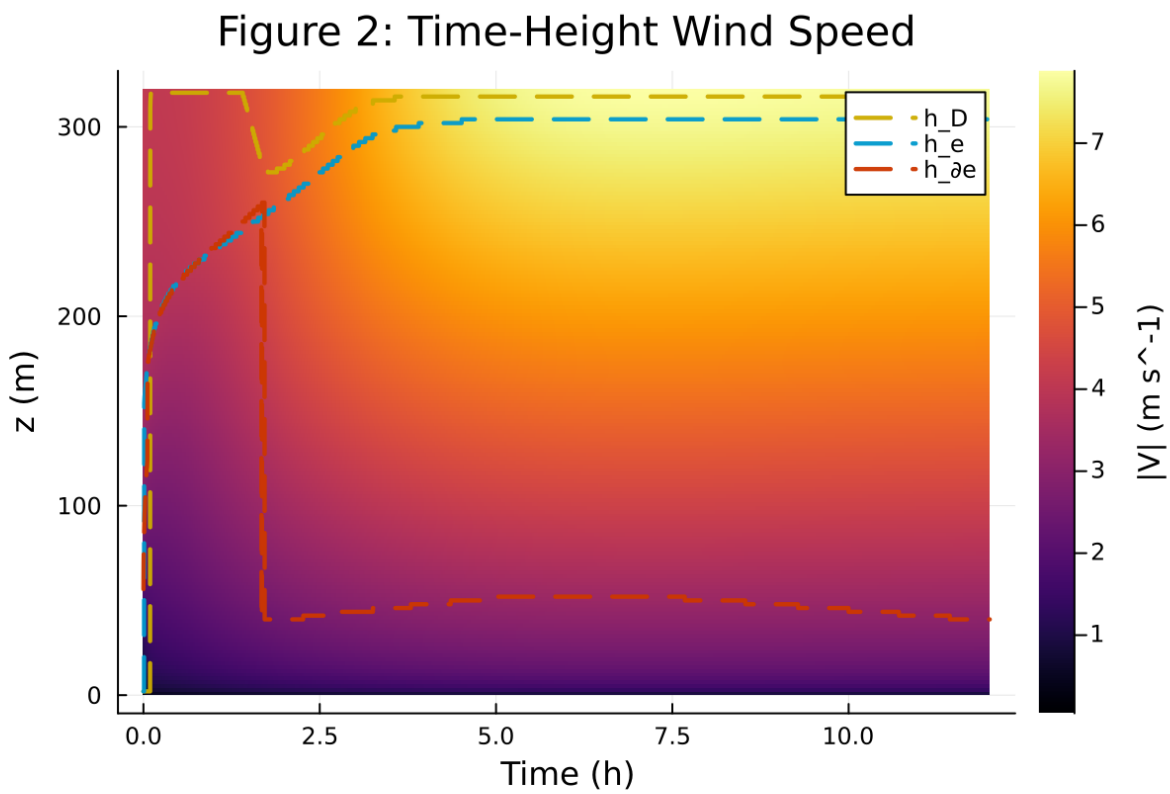


Figure 37: Time-height contour of wind speed with LLJ evolution

# 12 Case 6: sheba 250x2m 12h report

# 13 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba_high_top_fd/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREC Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success` .

## 13.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba_high_top_fd`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 152 / 1
- RHS evaluations: 115682
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

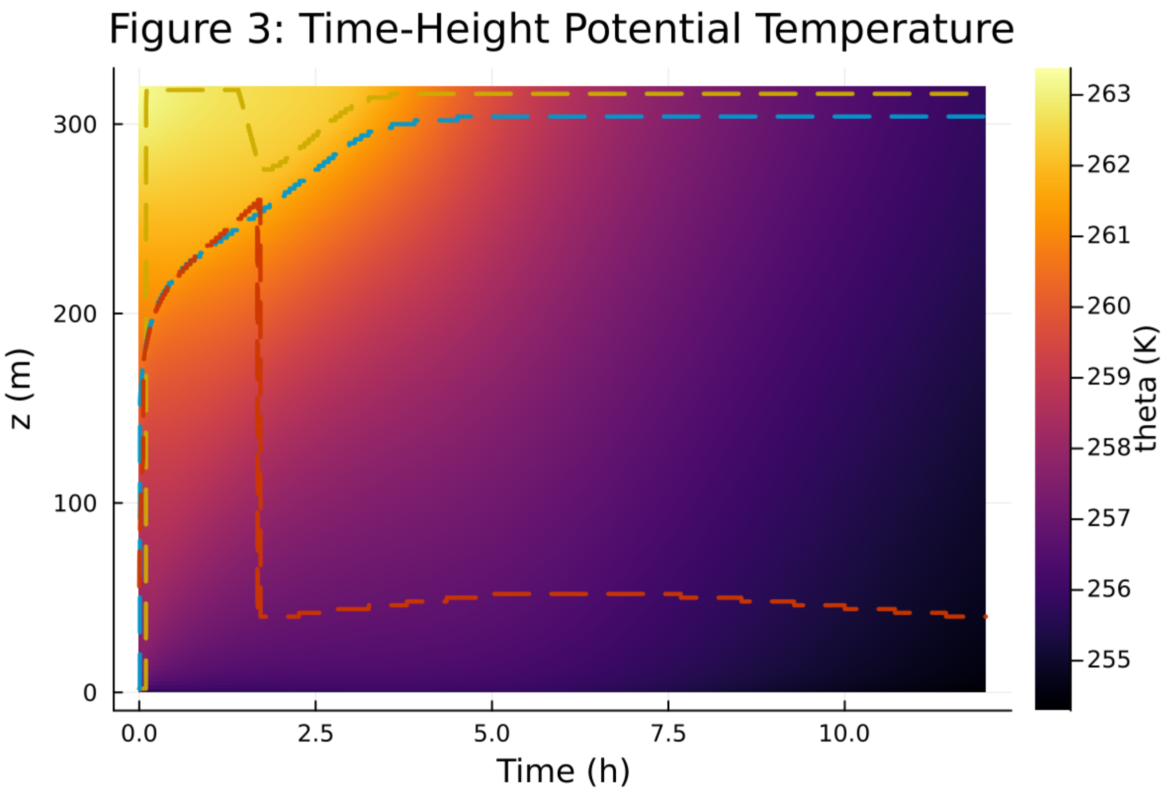


Figure 38: Time-height contour of potential temperature and inversion growth

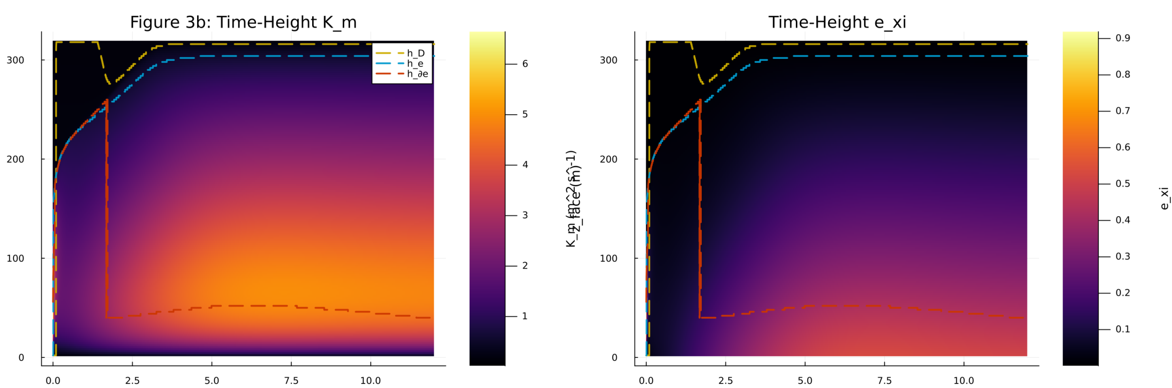


Figure 39: Vertical profiles of U, theta, and Km at representative times

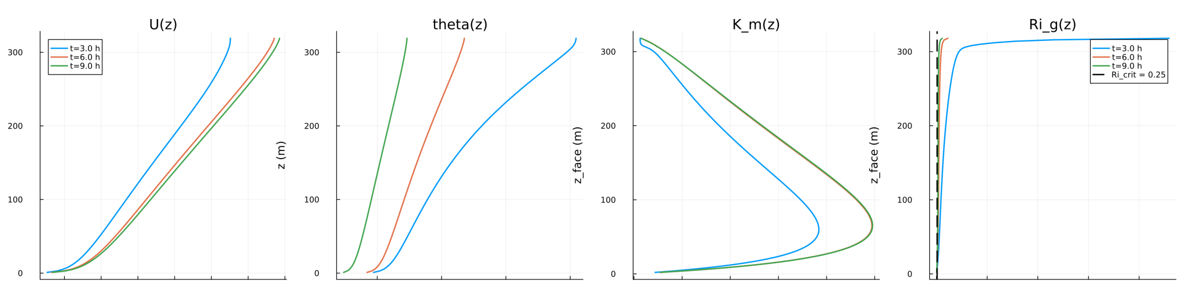


Figure 40: Surface energy budget components (R_n, H, G, storage)

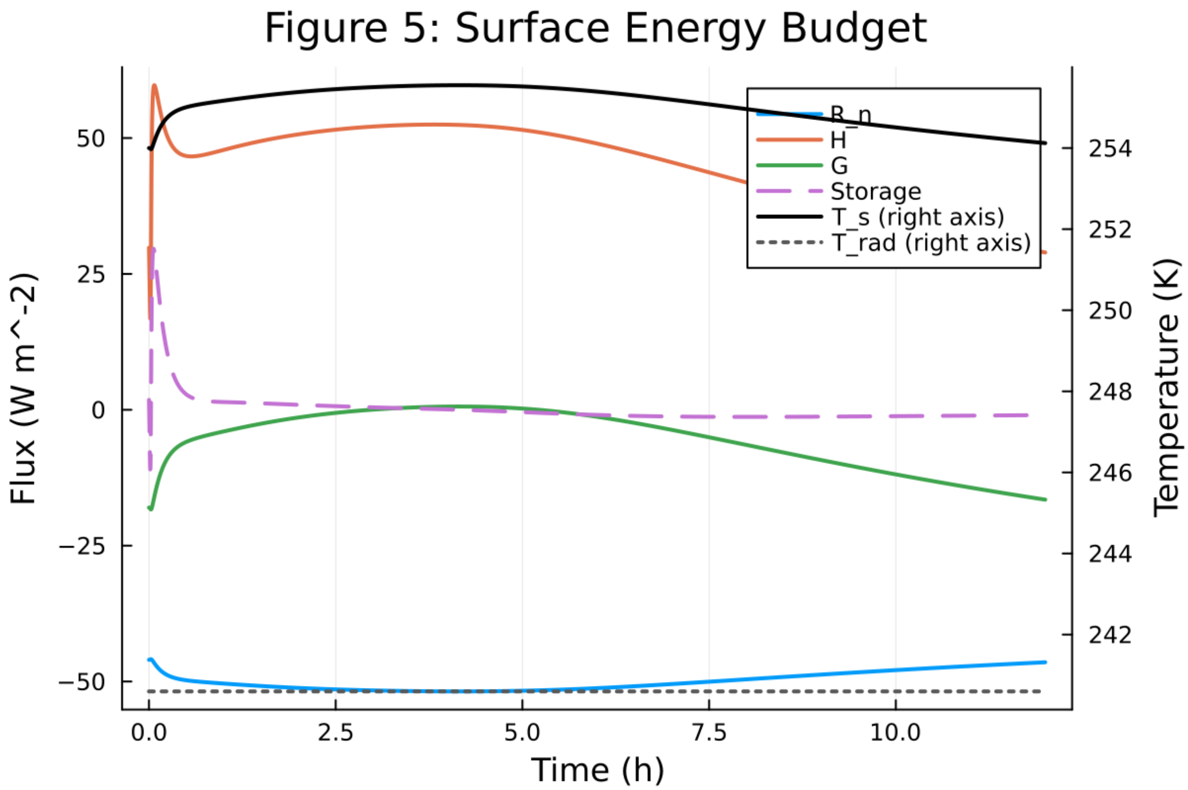


Figure 41: Phase portrait on regularized manifold (Delta vs e__xi)

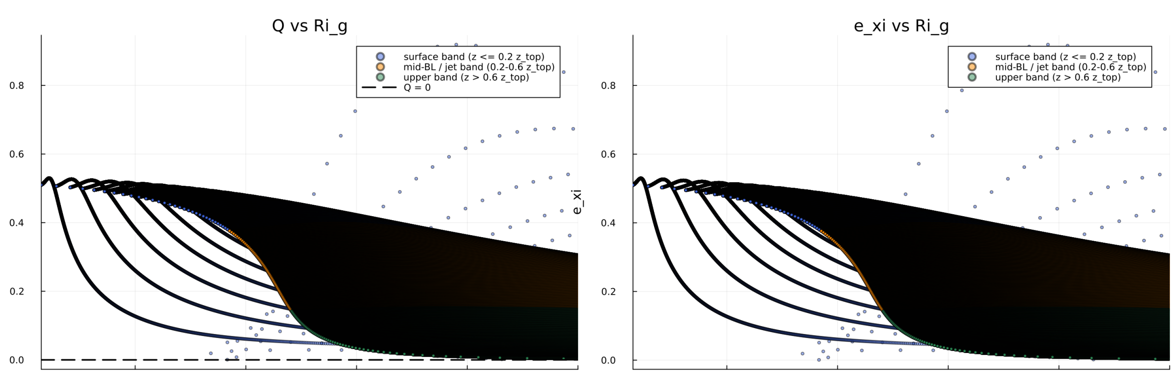


Figure 42: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

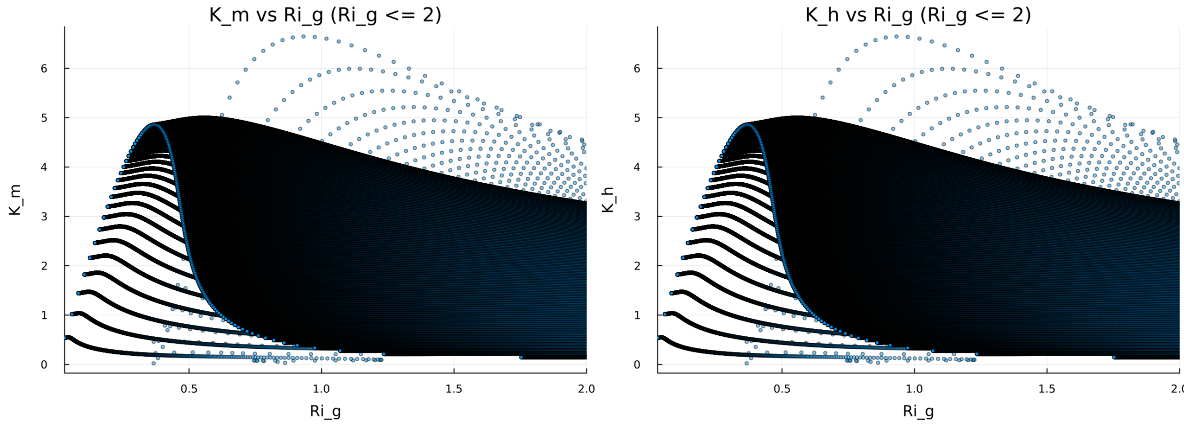


Figure 43: Fold proximity diagnostics versus time

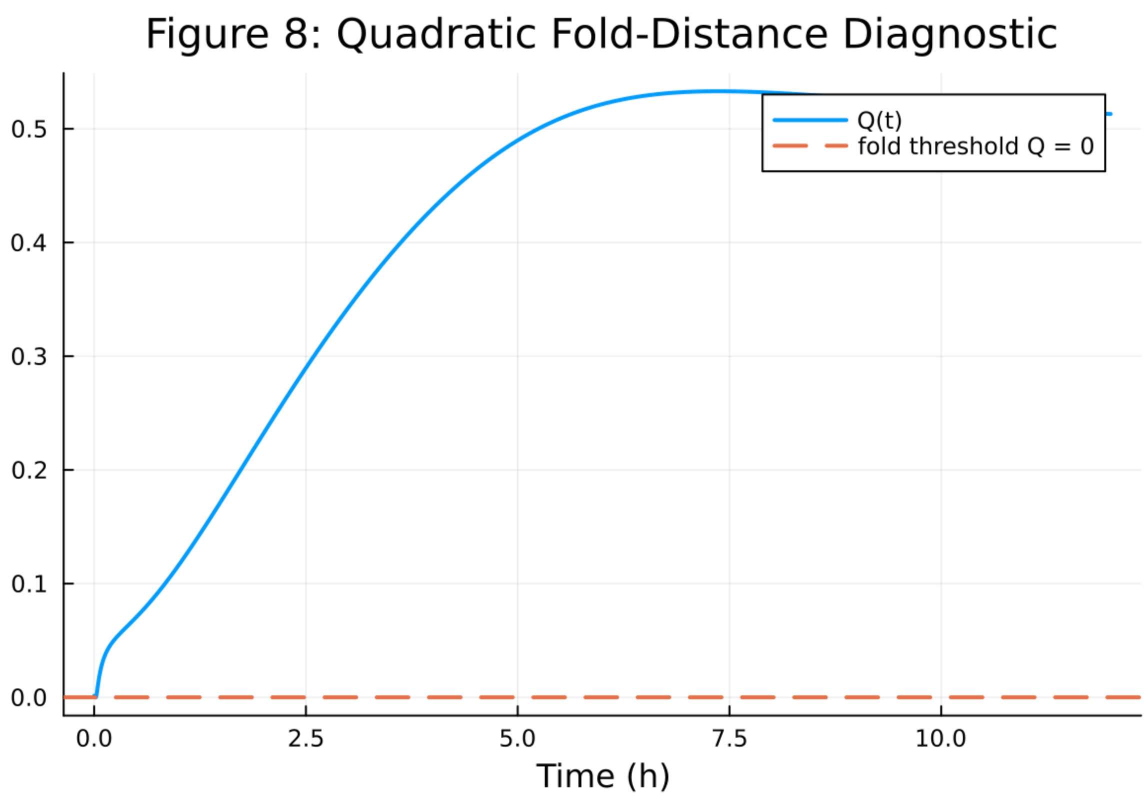


Figure 44: fig08 fold proximity.png

Table 6: Forcing and Boundary Condition Parameters

| **Parameter** | **Symbol** | **Value** |
|---|---|---|
| Zonal geostrophic wind | $U_g$ | 7.000000 m s$^{-1}$ |
| Meridional geostrophic wind | $V_g$ | 0.000000 m s$^{-1}$ |
| Surface roughness (momentum) | $z_{0m}$ | 1.000e-04 m |
| Surface roughness (heat) | $z_{0h}$ | 1.000e-05 m |
| Coriolis parameter | $f$ | 1.000e-04 s$^{-1}$ |
| Reference potential temperature | $\theta_a$ | 257.000000 K |
| Deep soil temperature | $T_{deep}$ | 255.500000 K |
| Downward longwave forcing | $R_\downarrow$ | 190.000000 W m$^{-2}$ |
| Soil conductivity proxy | $\lambda_s$ | 1.200000 |
| Soil slab depth | $d_{soil}$ | 0.100000 m |
| Surface diffusivity floor | $k_{\min,\mathrm{surf}}$ | 0.001000 m$^2$ s$^{-1}$ |
| Top thermal BC type | — | neumann |
| Top boundary potential temperature | $\theta_{top}$ | 257.000000 K |
| Top relaxation coefficient | $\lambda_{top}$ | 0.000000 s$^{-1}$ |

### 13.2 Forcing and Boundary Conditions

### 13.3 Numerical Verification

- Maximum surface energy closure error: 0 W m$^{-2}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.023796, 7.535015]$ m$^2$ s$^{-1}$
- Richardson number bounds: $Ri \in [0.030581, 3.273 \times 10^4]$

### 13.4 Generated Artifacts

- Time-series CSV: `results/sheba_high_top_fd/time_series.csv`
- Diagnostic summary JSON: `results/sheba_high_top_fd/summary.json`
- Payload JLD2: `results/sheba_high_top_fd/payload.jld2`

### 13.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

### 13.6 Included Plots

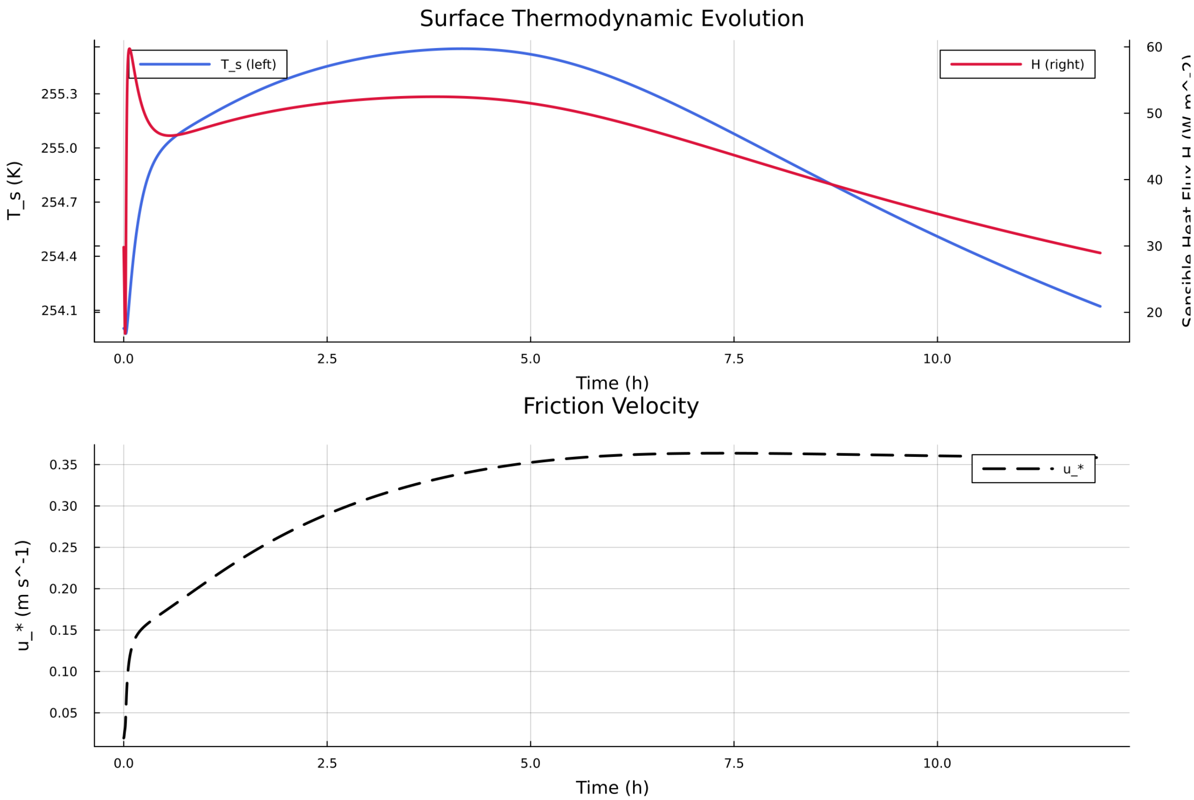


Figure 45: Time series of T__s, sensible heat flux, and friction velocity

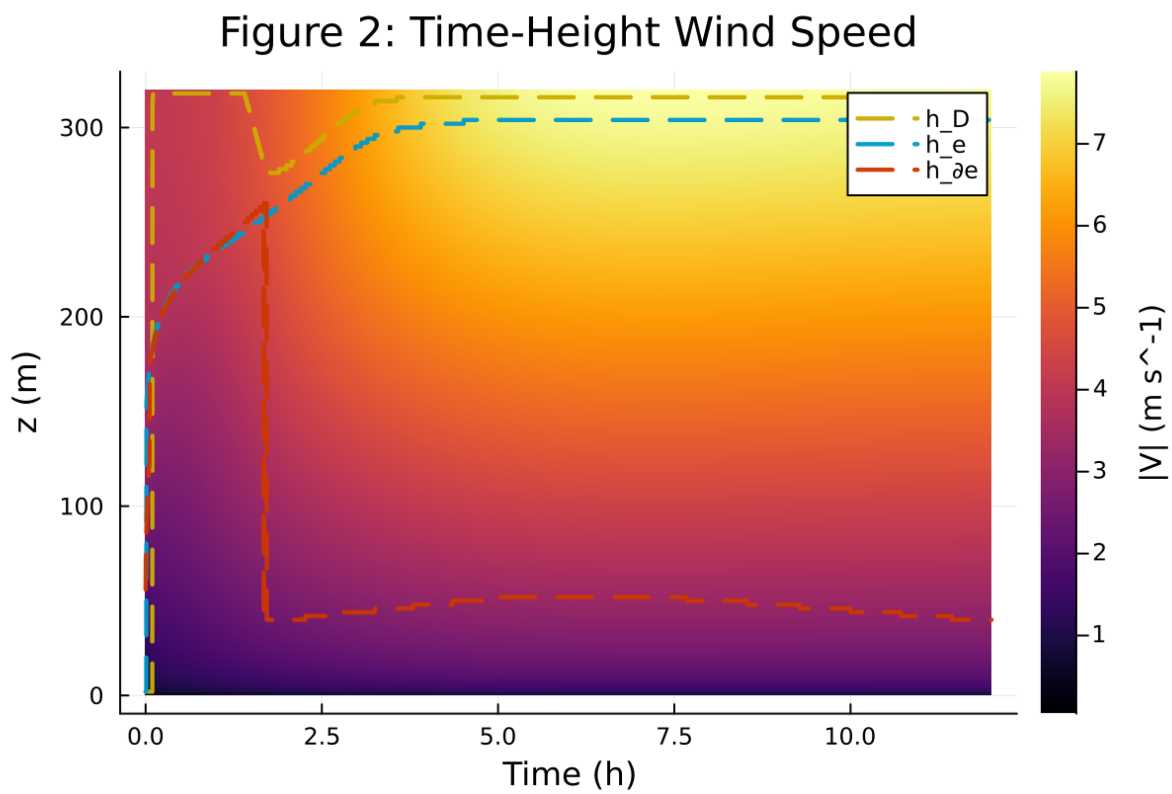


Figure 46: Time-height contour of wind speed with LLJ evolution

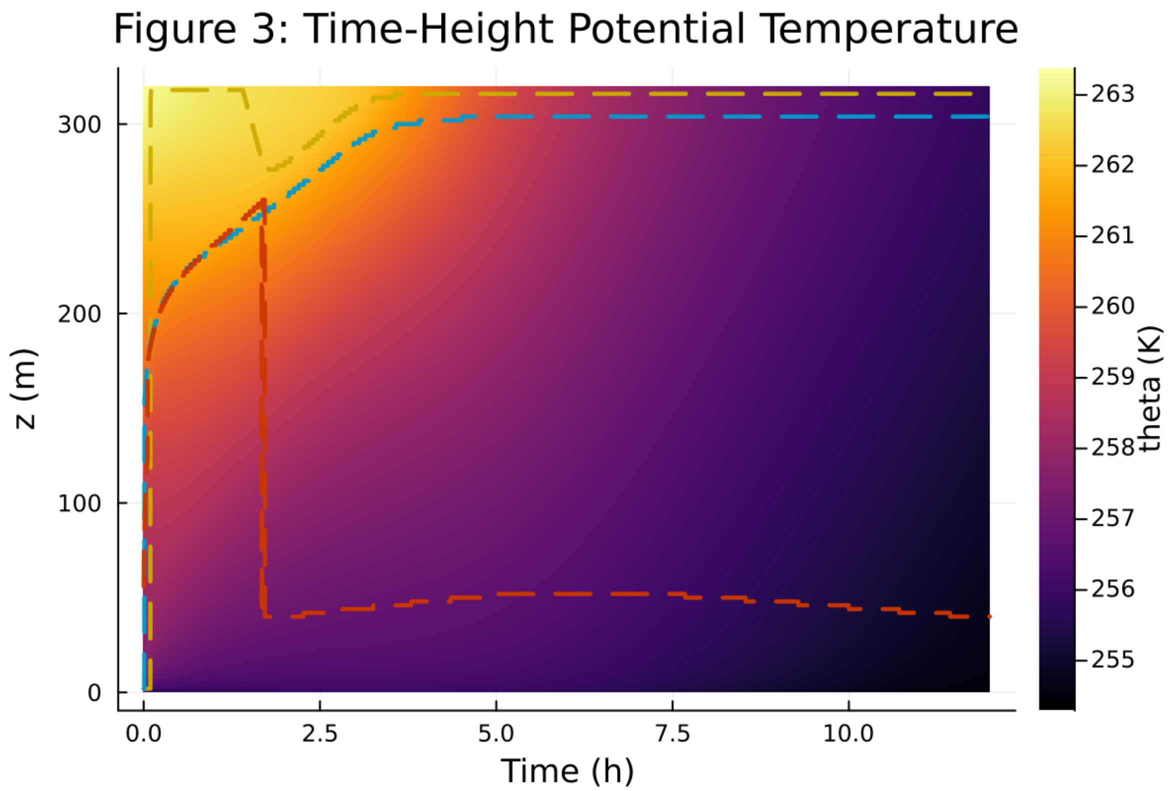


Figure 47: Time-height contour of potential temperature and inversion growth

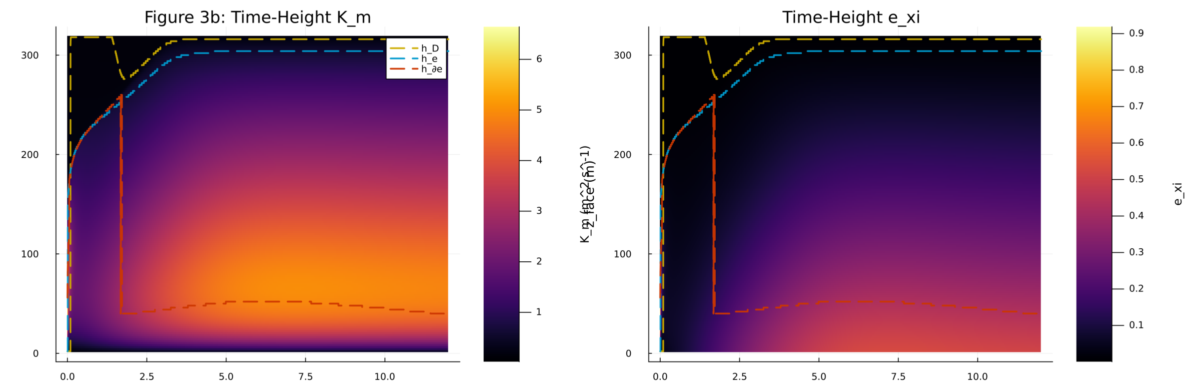


Figure 48: Vertical profiles of U, theta, and Km at representative times

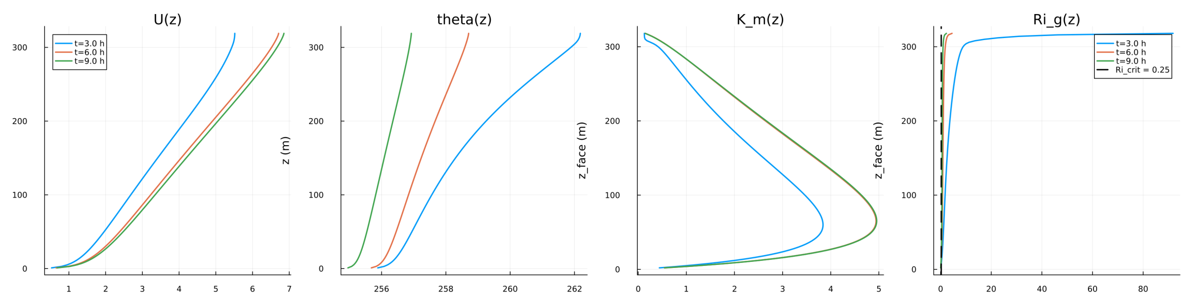


Figure 49: Surface energy budget components (R_n, H, G, storage)

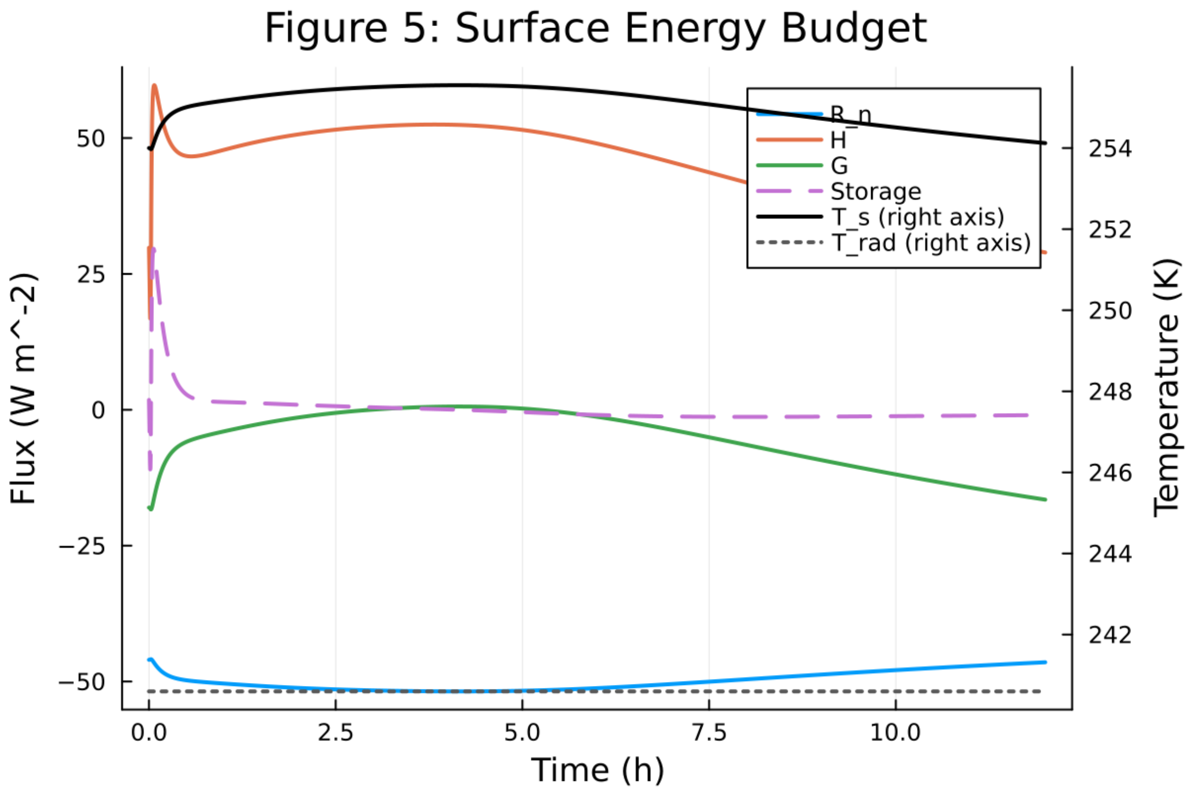


Figure 50: Phase portrait on regularized manifold (Delta vs e__xi)

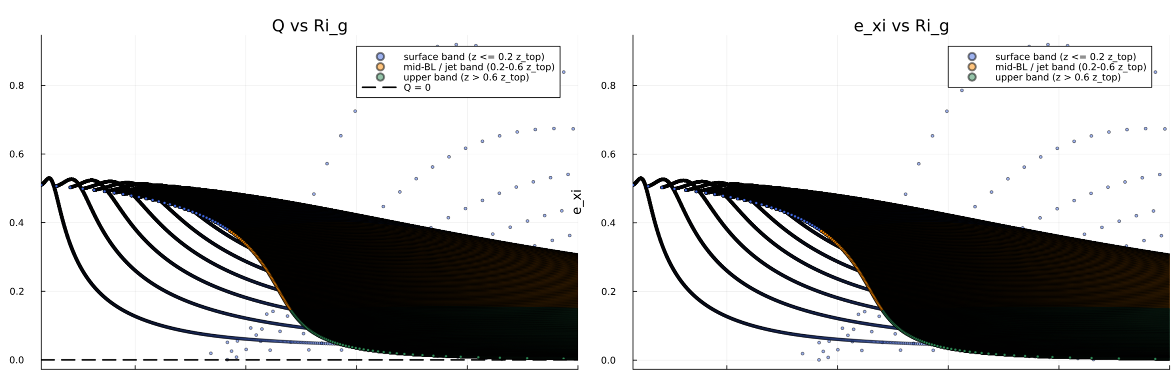


Figure 51: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

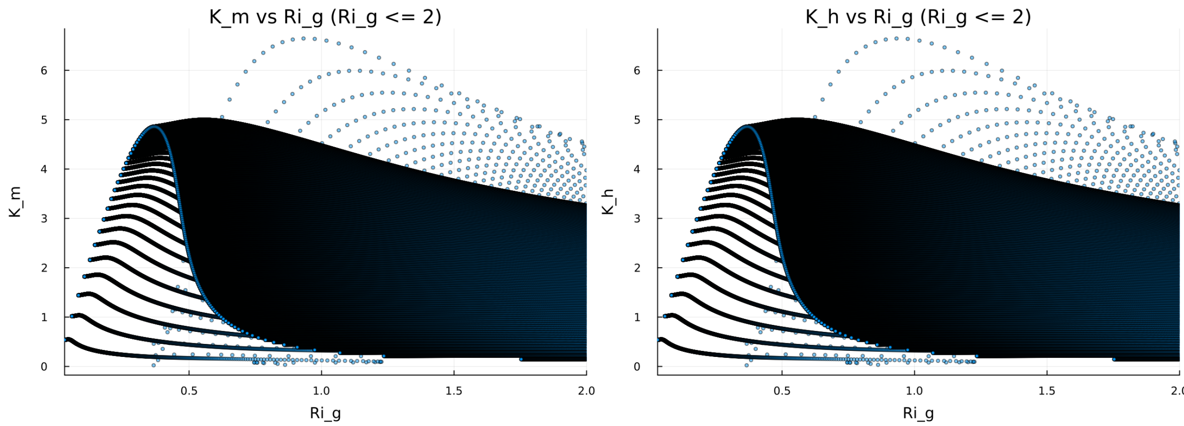


Figure 52: Fold proximity diagnostics versus time

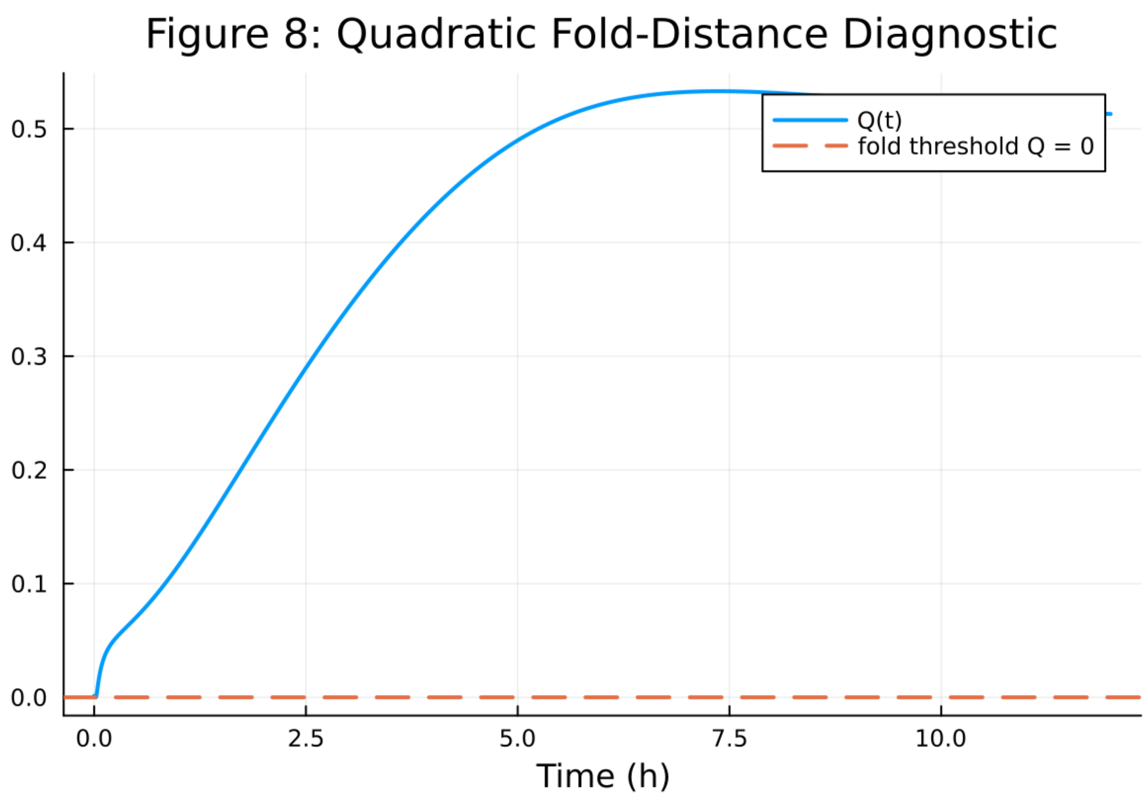


Figure 53: fig08 fold proximity.png